Disentangling Age, Time, and Cohort Effects in Lifecycle Income: A Proxy Machine Learning Approach

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October 17, 2026

Abstract

Two key facts economists have emphasized about the lifecycle income process are rapid growth of inequality and the average level of income over the lifecycle. However, empirical estimates of these facts are plagued by the difficulty of distinguishing between age, time, and cohort trends. Foundational work on the lifecycle profile of income inequality assumes that "time effects" are zero (Deaton and Paxson, 1994). But this strong assumption is rejected by the data. We propose a new proxy variable machine learning approach to the age-time-cohort problem. For the variance of income over the lifecycle, our method yields both a significantly smaller slope of the age profile for the U.S., and much less erratic age profiles for 11 other countries. For the average level of income over the life cycle, our method yields a larger slope for the U.S., and confirms earlier estimates showing that the slope of the income profile increases with economic development.

JEL Classification: E20, J20

*We thank Adrien Auclert, Avi Feller, Lihua Lei, Ben Moll, Jesse Rothstein, Pedro H.C. Sant'Anna, Stefan Wager and participants at seminars at various institutions for incredibly helpful comments and discussions. We thank the Alfred P. Sloan Foundation, and the Smith Richardson Foundation for financial support.

1 Introduction

The lifecycle income process for individuals is a key determinant of a wide range of macroeconomic phenomena. The slope of this process provides insight into human capital accumulation over the life-cycle and the incentives households face to save or borrow to smooth their consumption. The age profile of the cross-sectional variance of income is informative about the size and persistence of uninsurable income risk faced by households, which in turn determines their degree of precautionary savings and the value of social insurance programs. Earlier research has argued that the age profile of the cross-sectional variance of income increases sharply with age in a number of countries (Deaton and Paxson, 1994) and that the age profile of average income is more steeply upward sloping in high income countries than low income countries (Lagakos et al., 2018).

Estimating age profiles of average income, or of the cross-sectional variance of income is, however, less straightforward than it might first appear. An important challenge in estimating these profiles is the so-called age-time-cohort problem. Age equals time minus cohort ($a = t - c$). This means that age, time, and cohort effects cannot be disentangled without an identifying assumption. Consider how the cross-sectional variance of income evolves for a particular cohort over time. Suppose income inequality increases for this cohort between 2000 and 2010 as the cohort ages from age 30 (say) to 40. Does this imply the age profile of inequality slopes upward (age effects)? Or does it arise from the fact that 2000 was a business cycle peak, while 2010 was close to a business cycle trough (time effects)?

The age-time-cohort identification problem arises in many contexts in economics and other social sciences where researchers are working with data on the life-cycle of individuals, households, firms, or other entities. Examples include studies of the life-cycle profiles of voting, fertility, mortality, health, firm growth, firm innovation, in addition to consumption, income, wages, and labor supply. In all of these cases, researchers are interested in separating age effects, from cohort and time effects. (For example, do voters become more conservative with age?) But since age is equal to time minus cohort, linear age, time, and cohort effects are not separately identified.

Importantly, in a model with additive age effects, time effects, and cohort effects, the three sets of effects are identified up to a linear rotation (Deaton, 1997; Schulhofer-Wohl, 2018). In other words, it is only the slopes of the age, time, and cohort effects that are unidentified absent further restrictions. This implies that non-linear age-time-cohort effects are identified. It also means that any identifying restriction a researcher makes can be thought of as pinning down the slopes of the

age, time, and cohort profiles.

Different parts of the literature address the age-time-cohort problem in different ways. One approach is to introduce seemingly "innocuous" identification assumptions such as fixing two individual years to have the same time effect (i.e., coarsen the time effects) or two (or more) cohorts to have the same cohort effect (coarsen the cohort effects). It is crucial to recognize, however, that these are not truly innocuous assumptions. Rather, they amount to implicitly strong assumptions about the slopes of the age, time, or cohort profiles. Papay and Kraft (2015) and Bell (2020) present examples of how such "innocuous" assumptions can drive the conclusions of empirical studies.[1]

In the literature on household income risk, the standard approach to addressing the age-time-cohort problem—dating back to Deaton and Paxson (1994)—has been to drop time effects entirely. This approach yields the strongly upward-sloping age profile of cross-sectional income variance (Deaton and Paxson, 1994; Storesletten, Telmer, and Yaron, 2004; Guvenen, Karahan, Ozkan, and Song, 2021). A large literature on heterogeneous agent models in macroeconomics has used these statistics to estimate the size and persistence of uninsurable income risk. A steeply upward-sloping profile of income variance as individuals age is interpreted as being caused by an accumulation of large and very persistent uninsurable shocks to income.

The assumption of zero time effects is, however, rejected by the data. The cross-sectional distribution of income responds strongly to recessions (Storesletten, Telmer, and Yaron, 2004; Guvenen, Ozkan, and Song, 2014). Moreover, the choice of methodology matters for the substantive conclusions. Figure 1 contrasts estimates of the slope of the age profile of income variance for two methods: the conventional approach of dropping time effects (orange squares) versus the alternative of dropping cohort effects (blue dots). The "unconventional" approach—which allows for business cycle factors—yields a slope that is roughly half as large as the conventional approach. This large difference was noted by Heathcote, Storesletten, and Violante (2005), but the subsequent literature has largely ignored this concern.

We propose a new approach to solving the age-time-cohort problem that combines proxy variables with (debiased) machine learning. Our jumping-off point is the "proxy variable" approach in which a subset of the age, time, and cohort effects are modeled as being a function of a set of observable "proxies" (Heckman and Robb, 1985; Winship and Harding, 2008). We choose to model the time effects in this manner. Rather than assuming that time effects are zero, we assume that they are a function of business cycle proxy variables. A key challenge with this approach—which

[1] Schulhofer-Wohl (2018) and Rothstein (2023) address the identification problem by focusing only on non-linear age, time and cohort effects. This is, however, insufficient in our application — the variance of persistent income shocks depends directly on the upward trend of the age profile.

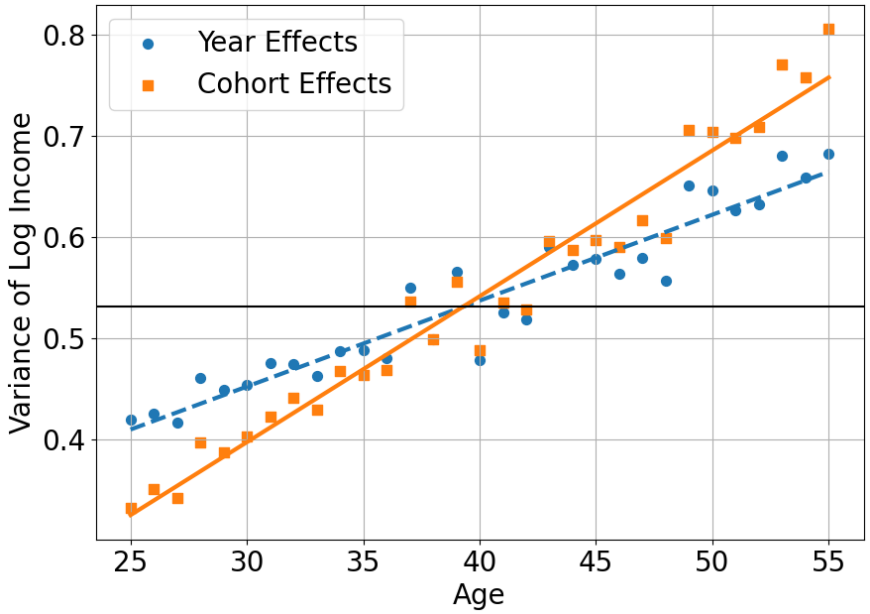


Figure 1: Two Estimates of the Age Profile of Income Variance

Note: The orange squares are age effects from a model that regresses the cross-sectional variance of log-income for a cohort at a points in time on age and cohort effects (but not time effects). The blue circles are age effects from an analogous model with age and time effects (but no cohort effects). We place a black horizontal line where the profiles intersect to aid visualization of the slopes. The data are from the PSID. The sample period is 1978-2019. See sections 2 and 3 for further details.

we demonstrate—is that small and seemingly-reasonable changes to the proxy model generate highly variable predictions for the age profile. The proxy variable approach, therefore, involves a non-trivial model selection problem.

Since machine-learning methods excel at model selection, we adopt the debiased machine learning approach of Chernozhukov, Newey, and Singh (2022) to estimate the slope of the age profile using proxy variables. We choose among a large number of proxy variables using cross-validation—a core idea from the machine learning literature. This approach yields a model that explains the data well, even out of sample, without us having to specify the precise functional form ahead of time. Furthermore, the debiasing procedure of Chernozhukov, Newey, and Singh (2022) yields asymptotically unbiased point estimates, valid standard errors, and an asymptotically normal confidence interval for the estimated slope of the age profile of cross-sectional income variance.

We apply our method to the estimation of the age profile of the cross-sectional variance of income and the average level of income using newly available data from the Global Repository of Income Dynamics (GRID) database on 12 countries including the US (as well as US data from the Panel Survey of Income Dynamics (PSID)). For the cross-sectional variance of income, our

debiased ML method yields much smaller estimates of the slope of the age profile for the United States than conventional methods. Moving from a model with age and cohort effects (but not time effects) to our debiased ML model decreases the slope of the age profile by about a factor of two. Interestingly, moving from PSID data to GRID data (which are much higher quality) also substantially reduces the slope of the age profile (by another factor of two). This implies a smaller role for persistent idiosyncratic shocks to income than are typically assumed in the literature.

Across countries, the conventional approach of dropping time effects yields extreme variability in the estimated slope of this age profile: for some countries the age profile is hugely positively sloped, while for others it has a large negative slope. In contrast, our debiased ML method yields slopes that clustered around modest positive values (similar to our estimate for the U.S.). These results suggest that time effects are important in explaining the evolution of the cross-sectional variance of income and ignoring them produces misleading estimates.

Our second application is to the age profile of average income. As before, conventional methods yield slopes of the age profile across countries that are implausibly widely dispersed. Our debiased ML model yields results that can more plausibly be interpreted through the lens of standard models of human capital accumulation. Moreover, our results provide substantive support for the principal finding of Lagakos et al. (2018) that the age profile of income is more steeply sloped in high-income countries than low income countries, even though we show that their favored identifying assumption (that the age profile is flat at the end of working life) is rejected in our data. Our analysis shows that controlling for cohort effects is particularly important for understanding the first moment of income growth over the lifecycle, while controlling for time effects is particularly important for the second moment.

From a methodological standpoint, there are two additional novel elements of our procedure that deviate from common practice in debiased machine learning and previous work on proxy variables. First, we implement cross-validation by leaving out whole blocks of contiguous years—a procedure we describe as "blocked cross-validation." Without blocking, we show that proxy variable methods can fail to achieve identification because flexible machine learning models are able to effectively recreate time trends.

Second, and related, we characterize a set of "forbidden proxies" that must be left out of our procedure to achieve identification. These are proxy variables that are so trend-like that they effectively encode the calendar year and therefore resurrect the original identification problem. In the appendix, we formalize this logic by deriving a lower bound on the variance with which any estimator can recover the age profile (the semiparametric efficiency bound). We show that

this variance bound deteriorates as the calendar year becomes more predictable from the proxies. The theoretical literature on proxy variable methods in the social sciences has not articulated this forbidden proxies concern, and papers like Winship and Harding (2008) include forbidden proxies and secure identification only through linearity in some cases. For the proxy variable literature to move toward more flexible methods, it is crucial to address this issue.

The remainder of the paper is organized as follows. Section 2 discusses the data on the life-cycle of income we use. Section 3 describes the conceptual framework we employ and reviews the age-time-cohort identification problem. Section 4 discusses the proxy approach and how it results in a non-trivial model selection problem. Section 5 introduces our debiased machine learning approach to solving this model selection problem. Section 6 presents our results on the age profile of the cross-sectional variance of income for the US and 11 other countries. Section 7 presents an analogous set of results on the age profile of the average level of income, as well as results on the relationship between the slope of this age profile and a country's level of economic development. Finally, section 8 concludes.

2 Data on Income

Our primary variables of interest are the mean and variance of the cross-sectional distribution of log male labor income by age and year. For the United States, we obtain estimates of these variables from two datasets. First, we use the Global Repository of Income Dynamics (GRID) (Guvenen et al., 2022a), a collection of aggregate statistics constructed using administrative micro-data from 12 countries. The data for the U.S. comes from the Longitudinal Employer-Household Dynamics Infrastructure files at the Census Bureau. The sample period is 1998-2019. We estimate our models using data on individuals of age 25-55. All statistics within GRID are constructed using a standardized set of sample selection criteria intended to harmonize the data across countries. For example, income is deflated by the PCE price deflator, income observations under a minimum wage-based threshold are removed. We discuss these sample selection criteria in more detail in Appendix A.

For comparison with earlier work, we also report results for the U.S. based on data from the Panel Study of Income Dynamics (PSID). Our sample period is 1978-2019. We calculate the mean and variance of log male labor income adjusted for changes in the PCE price deflator. We adopt a set of sample selection criteria meant to mimic the harmonized sample selection criteria used in the GRID data.

Beyond the U.S., GRID also records the mean and variance of log male income at the age-year level for 11 other countries: Argentina, Brazil, Canada, Denmark, France, Germany, Italy, Mexico, Norway, Spain, and Sweden. Data for these countries are taken from a range of administrative sources including tax and social security records. The sample period differs slightly across countries. See Appendix A for additional information.

We use a number of other data series as proxy variables for time effects. We discuss those data later in the paper.

3 How Big Is the Permanent Component of Lifecycle Income Risk?

Consider the following simple specification for the log income of individual i at age a:

$$\log Y_{i,a} = g(x_{i,a}) + y_{i,a} \tag{1}$$

$$y_{i,a} = \alpha_i + z_{i,a} + \epsilon_{i,a} \tag{2}$$

$$z_{i,a} = \rho z_{i,a-1} + \eta_{i,a}. \tag{3}$$

Log individual income $\log Y_{i,a}$ consists of a deterministic component $g(x_{i,a})$ and a stochastic component $y_{i,a}$. The deterministic component $g(x_{i,a})$ captures predictable variation in income with age. In section 7, we address the age-time-cohort problem in estimating this component. In this section, however, we focus on income *risk* over the lifecycle arising from the stochastic component $y_{i,a}$, which is not affected by $g(x_{i,a})$.

The stochastic component of log individual income $y_{i,a}$, consists of three parts: an individual fixed effect α_i, a persistent component $z_{i,a}$, and a transitory component $\epsilon_{i,a}$. The persistent component follows an AR(1) process with persistence ρ and innovations $\eta_{i,a}$. We denote the variances of α_i, $\epsilon_{i,a}$, and $\eta_{i,a}$ as $\sigma^2_\alpha, \sigma^2_\epsilon$, and σ^2_η, respectively, and write the initial variance of the persistent component as $\sigma^2_{z,0} \coloneqq \text{Var}[z_{i,0}]$. (Since mean income by age is $g(x_{i,a})$, $y_{i,a}$, $z_{i,0}$, α_i, $\epsilon_{i,a}$, and $\eta_{i,a}$ are all mean zero.)

A key question in models with uninsurable income risk is: what is the relative volatility of the persistent and transitory income shocks $\eta_{i,a}$ and $\epsilon_{i,a}$? Persistent income shocks are much harder to self-insure against. They therefore call for more precautionary savings and imply higher marginal propensities to consume for a given level of liquid assets. Deaton and Paxson (1994) point out that there is a straightforward way to identify the importance of persistent shocks in terms of the age profile of the cross-sectional variance of income. For expositional simplicity, consider the case of

$\rho = 1$, i.e., the case where the persistent component is a permanent component. In this case,[2] the variance of the stochastic component of log income is

$$\begin{aligned}\sigma_a^2 := E[y_{i,a}^2] &= \sigma_\alpha^2 + E[z_{i,a}^2] + \sigma_\epsilon^2 \\ &= \sigma_\alpha^2 + \sigma_{z,0}^2 + a\,\sigma_\eta^2 + \sigma_\epsilon^2. \end{aligned} \tag{4}$$

Notice that σ_a^2 is linear in age with a slope of σ_η^2. In other words, the slope of the age profile of income variance identifies the variance of the persistent component of income σ_η^2 when $\rho = 1$. The intuition for this is straightforward: the cross-sectional variance of income will fan out with age since the permanent shocks cumulate as people age (while transitory shocks do not). If $\rho < 1$, the age profile of cross-sectional income will be concave rather than linear. In that case, the degree to which it rises and the degree to which it is concave will identify σ_η^2 and ρ. As we saw in Figure 1, the age profile of the cross-sectional variance of income is approximately linear in the PSID. However, the larger sample size available from the GRID data yields a more precise estimate of the age profile that indicates that it is actually somewhat concave (see section 6).[3]

The challenge in using the age profile of the cross-sectional variance of income to identify the variance of persistent shocks to income is that the pure age profile is not directly observable in the data. We can compare the cross-sectional variance of income of people of different ages at a given point in time. But differences may then be due to the people belonging to different cohorts. We can follow the cross-sectional variance of income of a particular cohort as it ages. But differences may then be due to the fact that observations at different ages are taken at different times.

In the data, we observe log income at age a in year t, and can construct moments $E[(y_{i,a}^t)^2] =: \sigma_{at}^2$. Figure 2 illustrates this using data from GRID. We would like to control for the effect of calendar year t and birth cohort $c = t - a$ to get moments σ_a^2 that are "timeless" and "cohortless" and can then be used to estimate σ_η^2 via equation (4). However, the collinearity of age, time, and cohort makes this tricky to do.

Consider the fixed effects model

$$\sigma_{at}^2 = \alpha_a + \beta_t + \gamma_c + \nu_{at}, \tag{5}$$

where α_a is a set of age fixed effects, β_t is a set of time fixed effects, γ_c is a set of cohort fixed effects,

[2] Assuming that $\alpha_i, z_{i,0}, \eta_{ia}, \epsilon_{ia}$ are mutually uncorrelated.

[3] The autocovariances of the stochastic component of log income are also informative about σ_η^2. In the $\rho = 1$ case, they are $E[y_{i,a}y_{i,a+h}] = \sigma_\alpha^2 + \sigma_{z_0}^2 + a\,\sigma_\eta^2$.

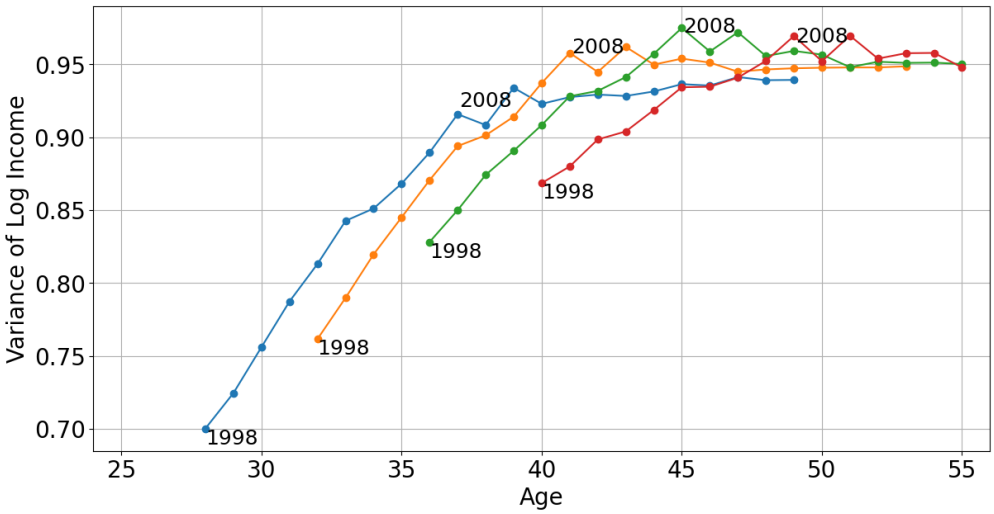


Figure 2: Cross-Sectional Income Variance by Age for Several US Cohorts

Note: Each point is the cross-sectional variance of log income in the US for a particular cohort in a particular year from the GRID data. The points for particular cohorts are connected by lines.

and ν_{at} an idiosyncratic component. If the three sets of fixed effects in this model were identified, α_a would be the pure age effects we seek to use to estimate σ_η^2. Unfortunately, α_a, β_t and γ_c are perfectly collinear since $a = t - c$. In particular, we can replace α_a, β_t and γ_c by

$$\tilde{\alpha}_a = \alpha_a + ka, \qquad \tilde{\beta}_t = \beta_t - kt, \qquad \tilde{\gamma}_c = \gamma_c + kc, \tag{6}$$

where k is any scalar and the predictions of the model will be unchanged. In other words, equation (5) is identified up to a linear rotation. This poses a particularly severe problem in our application since it is the slope of the age profile that identifies σ_η^2 (when $\rho = 1$). The slope is completely unidentified.

Any restriction that pins down k in equations (6) will render model (5) identified. Even minimal-seeming restrictions—such as setting two time effects equal to each other—will do the trick. It is tempting to view such restrictions as innocuous. In fact they are anything but innocuous since they determine the slope of all three sets of effects α_a, β_t and γ_c.

The standard approach in the literature on the age profile of the cross-sectional variance of income is to make the strong assumption that all time effects are zero, $\beta_t = 0\ \forall t$ (e.g., Deaton and Paxson, 1994; Guvenen et al., 2021). It is easy to see from Figure 2 that this assumption is rejected by the data. The cross-sectional variance of all cohorts bends down around 2008. This business cycle fluctuation cannot be captured by a combination of age and cohort effects. The timing of the bend is a non-linear time effect, which *is* identified as emphasized by Schulhofer-Wohl (2018).

4 Proxying for Time Effects

The identification problem discussed above implies that more structure is needed to identify a pure age profile of cross-sectional income variance σ_a^2. How should we formulate a compelling set of additional assumptions? The "proxy variable" approach responds to the need for additional structure by modeling one set of effects (age, time, or cohort) as a function of observable variables. An early formalization appeared in Heckman and Robb (1985), which was substantially generalized in Winship and Harding (2008).[4]

We propose to model the time effects in our application as being a function of observable variables Z_t. With this assumption (and maintaining linearity for expositional simplicity), equation (5) becomes

$$\sigma_{at}^2 = \alpha_a + \gamma_c + \beta^\top Z_t + \nu_{at}, \tag{7}$$

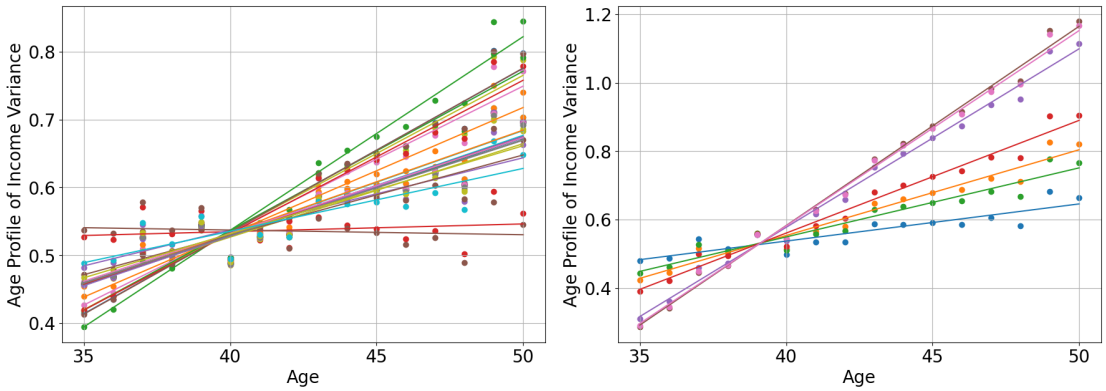
where Z_t is a vector of variables and β is a vector of coefficients. As long as the proxy variables Z_t are: (1) not perfectly linear in time, (2) capture all time effects, and (3) do not themselves depend on both age and cohort, all of the parameters of this model are identified. We describe the formal criteria for identification in Appendix B.

While the proxy variable approach can secure identification, its use raises an important model selection problem. There are many plausible proxy variables and there are many plausible functional forms. How should one choose among these? It turns out that this choice is quite consequential since results from proxy variable modeling of age, time, or cohort effects can be highly sensitive to these choices (see, e.g., Lu and Luo, 2021).

This sensitivity is easy to illustrate in our application. Figure 3 plots the profile of age effects (i.e., the α_a's) and their corresponding best fit lines that results from fitting the specification (7) with different plausible macroeconomic variables in Z_t (left panel) and different functional forms (right panel). Considering first the left panel: different choices of proxy variables can result in a slope of the age profile anywhere from slightly negative to strongly positive. In the right panel, we use a fixed set of proxies — unemployment, log GDP, and inflation — but consider polynomial expansions of the proxies of different degrees. As the polynomial degree we include sweeps from one to seven, the estimated slope ranges from relatively flat to strongly positive.

Figure 3 demonstrates that different seemingly-reasonable choices of proxy variables and func-

[4]Proxy variables have been used in many age-time-cohort applications, including to the level of consumption and income (Gourinchas and Parker, 2002), wealth holdings (Kapteyn et al., 2005), health (Portrait et al., 2010), female labor force participation (Euwals et al., 2011), and happiness (Su et al., 2022).



(a) Different Proxies

(b) Different Functional Forms

Figure 3: Sensitivity of Estimates of the Age Profile of Cross-Sectional Income Variance

Note: The left panel presents estimates of the age profile using (7), but using different proxies Z_t for time effects. We present estimates for 30 different proxy variables, used one at a time. The proxy variables are the 10 variables discussed in Section 5, the lag of each of these, and the first difference of each of these. We plot both the estimated age profiles (as dots) and the corresponding best-fit lines. The right panel presents estimates of the age profile with different functional forms for the model of time effects. In all cases, $Z_t = \{\text{Unemployment}, \log \text{GDP}, \text{Inflation}\}$, but we consider polynomial expansions of these proxies of different degrees from one to seven.

tional forms yield radically different estimates for the age slope. Clearly, our results will hinge crucially on how we resolve the model selection problem the proxy variable approach gives rise to. We now turn to that issue.

5 Our Methodology: Debiased Proxy Machine Learning

To solve the model selection problem discussed in section 4, we leverage several ideas from the machine learning literature. We consider a large set of candidate models with different proxy variables and different functional forms, including models with flexible interactions between age, cohort, and the proxies for time effects. We employ a blocked cross-validation procedure to choose among these models. Finally, we debias the resulting estimates of the slope of the age profile to obtain valid standard errors.

5.1 Proxies, Features, and Models

The specific machine learning methods we use are guided by the relatively modest size of our dataset. Many machine learning methods are designed for large datasets. Our datasets have between 434 and 806 observations (age-time pairs). For datasets of this size, ridge regression, Lasso, and kernel ridge regression perform well, while more flexible models like random forests and neural networks typically fail to generalize. We therefore use ridge regression, lasso, and

kernel ridge regression (KRR) as our basic set of models — we consider more complex models in Appendix D.2.

Our baseline set of proxy variables are ten macroeconomic time series: log GDP, log consumption, log investment, export values, log industrial production, male unemployment, log CPI, a short term interest rate, the exchange rate, and oil prices. For the US, we also include a measure of the skill premium. We summarize the particular data series we use, and their sources in Appendix A.4. For all variables, we consider eight transformations (in addition to the variable itself): the first difference, lag, square, square root, square of first difference, square root of first difference, square of lag, and square root of lag. All in all, this makes for $9 \times 10 = 90$ "features" (99 for the US).

Flexible machine learning models can theoretically learn the relevant features from the data. In practice, however, preprocessing and trimming of the feature set can make the learning task easier, substantially improving predictive performance. Given the modest size of our dataset, we are able to train our models with many different feature subsets. However, doing so for every combination of feature sets (2^{90} feature sets) is infeasible. We therefore perform two feature selection steps prior to training our models.

First, we remove proxies that trend too strongly with time as these threaten identification (see Appendix C.3). Specifically, we regress each feature on time and drop features for which this regression has an $R^2 > 0.75$. Second, we filter for proxies that have the strongest univariate relationship with the outcome variable. In particular, we fit kernel ridge regression models using the outcome variable and one proxy at a time.[5] As candidate features, we keep only the ten proxies with the smallest cross-validated mean squared error.

We train our models on all combinations of these ten chosen candidate features. For each sublist, we also include either age and cohort trends or age and cohort dummies. We therefore train our models on $2 \times (2^{10} - 1) = 2046$ different feature matrices for each country. For each method, we consider a range of possible values of the most relevant hyperparameters: the regularization penalties for lasso and ridge regressions, and the kernel and bandwidth for KRR. See Appendix D.1 for details. We normalize each feature (including the dummy variables) by subtracting their mean and dividing by their standard deviation.

[5] We use cross-validation to choose the kernel, kernel bandwidth, and penalty parameter for each proxy.

5.2 Model Selection with Blocked Cross-Validation

We use cross-validation—i.e., pseudo-out-of-sample fit—to choose among the many models discussed above. Standard cross-validation is performed by *randomly* splitting observations in the dataset into k disjoint "folds" (i.e., subsets). A typical approach is to "train" (i.e., estimate) each model on data from $k-1$ of the folds, and evaluate performance in terms of mean-squared error (MSE) on the remaining fold. This procedure is repeated using all possible splits into training and test sets. The overall performance of the model is then the average test MSE across splits and the model with the smallest cross-validated MSE is chosen.

This plain-vanilla cross-validation procedure runs into "data leakage" problems when the data are dependent as is the case in our panel data setting. Recall that our data are indexed by age and time. We therefore have many observations from each year (one for those aged 35, another for those aged 36, etc.). The standard random split of the data into k folds will typically result in data from a given year being divided across folds, and therefore both training and test data can simultaneously contain observations from the same year. As a consequence, the model is never evaluated out-of-sample when it comes to time. The problem then is that the flexible models we employ can exploit the dependence between training and test sets to spuriously overfit the data. Since business cycles unfold over several years, dependence between observations close in time is also a concern.

To mitigate data leakage between training and test sets due to dependent data, we use *blocked cross-validation*. Rather than splitting the observations randomly, we partition the sample into evenly-sized blocks of contiguous years.[6] By default, we use three blocks as illustrated in Figure 4. To reduce boundary effects, we choose the starting year of the partitions randomly, and allow the blocks to wrap around from the end to the start of the sample. Finally, to reduce variance, we perform repeated cross-validation (Burman, 1989; Kim, 2009), where we repeat the procedure over many random partition splits and then take the average. Repeated cross-validation is similarly recommended in Chernozhukov et al. (2018).

Blocked cross-validation plays an especially important role in our setting. Dependence between observations can degrade the quality of model selection to differing degrees in different

[6]There is a fairly extensive literature — both in machine learning and econometrics — going back to the late 80's on cross-validation variants adapted for dependent data. Influential early work includes Snijders (1988); Györfi et al. (1989); Burman et al. (1994); Racine (2000). A modern empirical literature using machine learning methods (Bergmeir and Benítez, 2012; Bergmeir et al., 2018; Cerqueira et al., 2020; Liu and Zhou, 2024) has consistently found that the blocked cross-validation method of Snijders (1988) performs well across a variety of different time-series settings. Racine (2000) additionally introduces "gaps" by dropping observations directly between train and test, but the modern empirical literature finds this does not improve performance in practice.

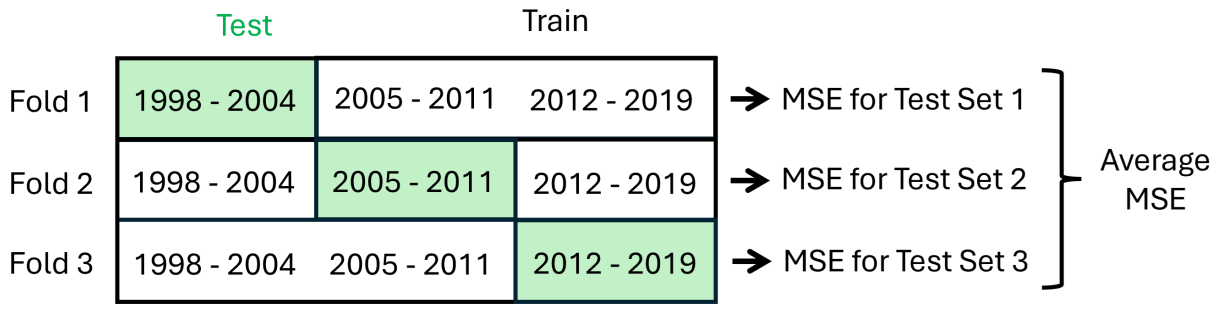


Figure 4: Our Blocked Cross-Validation Procedure with Three Blocks

settings — compare Bergmeir et al. (2018) and Liu and Zhou (2024). Surprisingly, in our setting, if we use plain-vanilla cross-validation for model selection, the slope of the age profile of σ_a^2 is *completely unidentified* even with proxy variables. We demonstrate this formally in Appendix C.3. In order to identify the contribution of the time effects, the proxy variables must be able to make predictions on unseen years.[7]

5.3 Computing the Slope of the Age Profile

The machine learning procedure described above yields a predictive model for σ_{at}^2 that takes as input age a, cohort c, and proxy variables z. We will write $\hat{m}$ to denote our trained model as a function of its inputs. For a given age-time pair (with implied cohort and proxies associated with that year), our model's prediction is $\hat{\sigma}_{at}^2 = \hat{m}(a, c, z)$.

Our primary object of interest is not the individual predictions $\hat{m}(a, c, z)$ but rather the slope of the predictions as age varies. In the familiar special case of a linear model (7) we have $\hat{m}(a, c, z) = \hat{\alpha}_a + \hat{\beta}^\top z + \hat{\gamma}_c$. In this case, the age effects are the $\hat{\alpha}_a$'s and one can regress these on age to get the slope of the age profile of $\hat{m}(a, c, z)$.

More generally, $\hat{m}(a, c, z)$ may be non-linear and it is not as obvious how to define age effects. We adopt a definition based on *counterfactually* changing age, but leaving cohort and time fixed. We implement this in practice for the age effect at age 35 (say) by making predictions for σ_{35}^2 using $\hat{m}$ for every observation in our dataset but counterfactually setting the age to 35 in all observations. In other words, we calculate the sample analog of $\hat{\sigma}_{35}^2 := E[\hat{m}(35, c, z)]$. Doing this for every age yields a set of age effects. We can then calculate the slope of the age effects by regressing these age effects on age. In the special case of the linear model (7), we recover the usual fixed effects

[7] As a special case, using the AIC or BIC for model selection also yields complete lack of identification. The AIC and BIC are asymptotically equivalent to leave-one-out and leave-k-out cross-validation respectively (Stone, 1977; Shao, 1997). Winship and Harding (2008) use the AIC for model selection, but they secure identification through a linearity assumption.

profile up to a constant. Specifically, we have $\hat{\sigma}^2_{35} = \hat{\alpha}_{35} + E[\hat{\beta}^\top z + \hat{\gamma}_c]$ where the expectation term is an intercept independent of age. Our non-parametric definition has the advantage that it can be computed for any machine learning model (whether or not the model uses a notion of age fixed effects internally). A similar definition using counterfactual averages was recently independently proposed for cohort profiles in Reynolds (2024).

5.4 Debiased Machine Learning and Standard Errors

Machine learning estimators virtually always use bias to reduce variance. For example, Ridge and Kernel Ridge regressions use regularization to explicitly shift their coefficients towards zero. Regularization can be optimal for minimizing out-of-sample mean square error (by preventing overfitting). But in our context—absent some adjustment—the bias will pass through to our estimate of the slope of the age effects. A key concern for us is therefore: will we estimate a smaller slope with our machine learning approach simply due to a mechanical effect of regularization?

We address this issue by using the "Debiased Machine Learning" (DML) approach of Chernozhukov et al. (2018, 2022). Recall that $\hat{m}$ is our cross-validated machine learning predictor of σ^2_{at}. Let slope$(\hat{m})$ be our estimate of the slope of the age profile as described in Section 5.3. The key idea behind DML is to estimate a function, $\hat{\alpha}(a, c, z)$, associated with the slope operator, slope$(\cdot)$, called its "Riesz representer." Chernozhukov et al. (2022) show that adding a bias correction term to our baseline estimate of the slope,

$$\text{slope}(\hat{m}) + \hat{E}\big[\hat{\alpha}(a, c, z) \cdot \big(\underbrace{\sigma^2_{at} - \hat{m}(a, c, z)}_{\text{prediction errors}}\big)\big], \tag{8}$$

results in an asymptotically unbiased estimate with a valid normal confidence interval, under minimal conditions. This holds even if the estimates $\hat{m}$ and $\hat{\alpha}$ are both biased themselves.

It remains to define and estimate the Riesz representer. Our main strategy for this is to adopt the "automatic" estimation strategy of Chernozhukov, Newey, and Singh (2022). Their approach uses the fact that the true Riesz representer is the unique minimizer of the following optimization problem:

$$\min_{\alpha} \big\{E[\alpha(a, c, z)^2 - 2 \cdot \text{slope}(\alpha)]\big\}. \tag{9}$$

We can estimate $\hat{\alpha}$ using machine learning algorithms by minimizing the loss (9). In doing this, we consider the same large set of candidate models and same set of proxy variables that we used

to estimate $\hat{m}$. Likewise, for model selection, we use the same blocked cross-validation procedure. A downside of characterizing the Riesz representer as the solution to the optimization problem (9) is that it provides little intuition. We attempt to provide a more insightful discussion of the Riesz representer in Appendix C.

With estimates of $\hat{m}$ and $\hat{\alpha}$, we can compute the debiased estimate of the slope using (8). Note, that to achieve valid inference, we should estimate $\hat{m}$ and $\hat{\alpha}$ in a separate sample from the one we use to compute (8). In practice, we use a cross-fitting procedure. The notation is cumbersome. So, we defer a complete description of the cross-fit estimator to Appendix D.5. Because we fit both $\hat{m}$ and $\hat{\alpha}$ using machine learning and then combine them, this procedure is sometimes called "double machine learning" or "doubly-robust estimation".[8]

Under minimal conditions derived in Chernozhukov et al. (2023), plugging our resulting $\hat{m}$ and $\hat{\alpha}$ into equation (8) results in an unbiased estimator of the slope. This implies that the smaller estimates of the slope our method yields compared to previous estimates are not driven mechanically by regularization bias. To further buttress this point, in Appendix G, we perform a simulation study with semi-synthetic data generation processes calibrated to the U.S. GRID data to assess the validity of our methodology. In the simulation study, we assume the ground-truth slope to be high as a "placebo" to determine whether our approach might generate an artificially low slope. We show this is not the case. Our debiased proxy machine learning estimator accurately recovers the high slope in simulation.

In addition to ensuring unbiasedness, the debiasing procedure also allows us to compute an asymptotically-normal confidence interval. We give a complete description of how we compute the point estimate and standard errors in Appendices D.4 and D.5. In our simulation study, we find that naively computing standard errors without debiasing results in a confidence interval that drastically undercovers: the naive 95% confidence interval contains the truth 0% of the time, whereas our debiased confidence interval contains the truth 94.7% of the time. Thus, debiased machine learning also plays a crucial role in enabling accurate uncertainty quantification.

We provide an outline of our end-to-end procedure in Algorithm 1. We emphasize that our estimator can be applied to other age-time-cohort settings by replacing σ^2_{at} with any other outcome variable. Our analysis in section 7 is a second application.

[8] This approach dates back to at least Robins et al. (1994) for the estimation of average treatment effects. See Kennedy (2022) for a review. For earlier work in the econometrics literature that takes advantage of writing the Riesz representer as the solution to an optimization problem, see Ai and Chen (2003).

Algorithm 1 Debiased Proxy Machine Learning with Repeated Blocked Cross-Fitting

INPUT: number of folds K, number of repetitions S, candidate predictive models $\mathcal{M} = \{m_1, \ldots, m_J\}$, and candidate weighting models $\mathcal{A} = \{\alpha_1, \ldots, \alpha_L\}$
for repetition $s = 1, \ldots, S$ **do**
  Draw an offset u_s uniformly from $\{0, \ldots, T-1\}$ and partition the T years into K disjoint circular blocks $\mathcal{T}_{s1}, \ldots, \mathcal{T}_{sK}$.
  for fold $k = 1, \ldots, K$ **do**
    Let $\mathcal{D}_{sk}^{\text{test}}$ be the observations in years $\mathcal{T}_{sk}$, and $\mathcal{D}_{sk}^{\text{train}}$ the remaining observations.
    for $m_j \in \mathcal{M}$ **do**
      Train m_j on $\mathcal{D}_{sk}^{\text{train}}$ to predict σ_{at}^2 given a, c, z; denote the fit $\hat{m}_{j,sk}$.
      Compute $\text{MSE}_{j,sk}$, the mean-squared error of $\hat{m}_{j,sk}$ on $\mathcal{D}_{sk}^{\text{test}}$.
    end for
    for $\alpha_\ell \in \mathcal{A}$ **do**
      Train α_ℓ on $\mathcal{D}_{sk}^{\text{train}}$ to minimize the Riesz loss (9); denote the fit $\hat{\alpha}_{\ell,sk}$.
      Compute $\text{RL}_{\ell,sk}$, the Riesz loss of $\hat{\alpha}_{\ell,sk}$ on $\mathcal{D}_{sk}^{\text{test}}$.
    end for
  end for
  Select $\hat{j}_s = \text{argmin}_j \frac{1}{K}\sum_{k=1}^{K} \text{MSE}_{j,sk}$ and $\hat{\ell}_s = \text{argmin}_\ell \frac{1}{K}\sum_{k=1}^{K} \text{RL}_{\ell,sk}$.
  Set $\hat{m}_{sk} \coloneqq \hat{m}_{\hat{j}_s,sk}$ and $\hat{\alpha}_{sk} \coloneqq \hat{\alpha}_{\hat{\ell}_s,sk}$ for $k = 1, \ldots, K$.
  Compute $\hat{\theta}_s$ as in (18) and $\hat{V}_s$ as in (19).
end for
Aggregate across repetitions: compute $\hat{\theta}$ and $\hat{V}$ as in (21), and $\widehat{\text{std err}}$ as $\sqrt{\hat{V}/n}$.
OUTPUT: $\hat{\theta}$ and $\widehat{\text{std err}}$

6 The Slope of the Age Profile of Income Inequality

We now employ the method laid out above to estimate the slope of the age profile of the cross-sectional variance of income. We first present results for the United States and then for the 11 other countries for which we have data in the GRID dataset.

6.1 Life-Cycle Income Inequality in the United States

Table 1 reports our estimates of the slope of the age profile of the cross-sectional variance of income for the United States. We present results from the GRID and PSID datasets. For the PSID, we present results for two sample periods: 1998-2019 (for comparability with our GRID results) and 1978-2019. We present results for four models: the conventional "age-cohort" model (which includes age and cohort fixed effects, but no time fixed effects), the "age-time" model (which includes age and time fixed effects, but no cohort fixed effects), and two versions of our machine learning (ML) model (with and without our debiasing procedure). For the debiased ML estimates, we include estimates of the 95% confidence interval of our estimate.

Table 1: Slope of Age Effects of the Cross-Sectional Variance of Income

Dataset	Years	Age-Cohort	Age-Time	Biased ML	Debiased ML
GRID	1998-2019	0.67	0.40	0.36	0.33 ± 0.13
PSID	1998-2019	1.34	0.64	0.56	1.08 ± 0.69
PSID	1978-2019	1.39	0.90	0.84	0.84 ± 0.35

Note: The table presents results on the slope of the age effects of the cross-sectional variance of log income for the United States. We present the slopes multiplied by 100 to ease readability. The "Age-Cohort" column presents results from a model with age and cohort effects, but no time effects. The "Age-Time" column presents results from a model with age and time effects, but no cohort effects. The "Biased ML" column presents results from our model without debiasing. The "Debiased ML" column presents results for our model including our debiasing procedure. This last column also presents estimates of the 95% confidence interval of the estimate. The PSID results use sample selection criteria similar to those for GRID.

We estimate these four models on data from individuals of age 25 to 55. However, to limit the influence of schooling and retirement, our main estimates of the slope of the age profile are for age effects over the age range 35 to 50. Later in this section, we present results on the age profile for the full age range. The model that is selected by our model selection procedure is a kernel ridge regression with age and cohort trends, the square root of age, the logarithm of the price of oil, the logarithm of the price of oil lagged, and squared log industrial production lagged.

A result that emerges from Table 1 is that our debiased ML estimate of the slope of the age profile is considerably smaller than the age-cohort estimate that has been widely used in the literature. For the GRID dataset, the debiased ML estimate of the slope is 0.33% per year, while the age-cohort estimate is 0.67% per year.[9] For the PSID with sample period 1978-2019, the debiased ML estimate is 0.84% per year, while the age-cohort estimate is 1.39% per year.

We conclude from this that the widely-used age-cohort estimate of the slope of the age profile of cross-sectional income variance likely yields estimates that are considerably upward biased. The age-cohort estimate is outside the 95% confidence interval for the debiased ML estimate in both of the cases just discussed. In these cases, our debiased ML estimate is much closer to the age-time estimate than the age-cohort estimate.

A second result that emerges from Table 1 is that we estimate a considerably smaller slope of the age profile with GRID data than with PSID data. The debiased ML estimate for GRID is 0.33% per year, while it is 1.08% per year and 0.84% per year for the two different sample periods we use for PSID. Our estimate for the GRID dataset is much more precise than our estimate for PSID. The 95% confidence interval for our GRID estimate is $[0.20, 0.46]$, while, even for the longer sample period in the PSID, our estimate is much wider at $[0.49, 1.19]$.

[9] In a slight abuse of notation, we write a slope of 0.33% per year to mean that the cross-sectional variance of log income rises by 0.0033 per year of age.

Table 2 presents estimates for an alternative measure of the growth in the cross-sectional variance of income over the life cycle: the change in the cross-sectional variance of income between age 35 and age 50. These estimates tell the same story as the slope estimates in Table 1. The debiased ML estimates are substantially smaller than the age-cohort estimates and the estimates from GRID are substantially smaller than the estimates from the PSID.

Table 2: Change in Cross-Sectional Variance of Income from Age 35 to 50

Dataset	Years	Age-Cohort	Age-Time	Biased ML	Debiased ML
GRID	1998-2019	10.4	6.5	5.7	5.3
PSID	1998-2019	18.9	8.4	9.6	13.1
PSID	1978-2019	24.0	15.8	14.7	14.7

Note: The table presents results on the change in the cross-sectional variance of income from age 35 to age 50. We multiply these estimates by 100 for readability. See the note for Table 1 for more detail.

Figure 5 plots our estimated age profiles for the GRID dataset. This estimated age profile is very smooth, reflecting the fact that the GRID dataset is based on a large administrative dataset. The estimated age profile for the PSID is estimated with vastly more sampling error (see Figure H.1 in the appendix). We can clearly see the difference in slope between the profile for the age-cohort model and the profile for the debiased ML model in Figure 5.

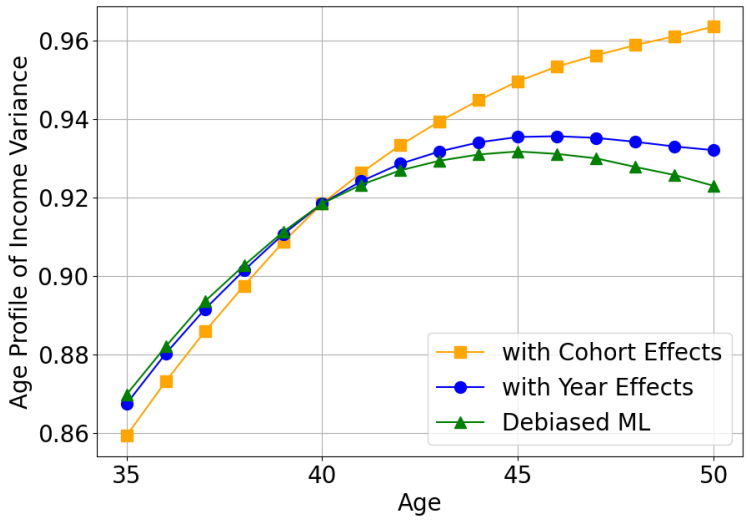


Figure 5: Age Profile of Cross-Sectional Income Variance in the US

Note: The figure plots the estimated age profile of the cross-sectional variance of income for three models using GRID data.

Another notable feature of the age profile in Figure 5 is that it is concave. This suggests that income shocks are less than fully permanent. Table 3 presents GMM estimates of the persistence parameter ρ for the income process presented in Section 3. (We describe the details of our GMM procedure in Appendix F.) For the GRID dataset, our debiased ML estimate of ρ is 0.854. This

estimate implies that the half-life of income shocks in GRID is about 4 years. As with our estimates of the slope of the age profile, our debiased ML estimate for ρ is smaller than the age-cohort estimate. The age-cohort estimate for the GRID dataset is 0.934 implying a half-life of 10 years. Our debiased ML estimate therefore implies not only a smaller slope of the age profile, but also less persistence of income shocks.

Table 3: Persistence of Income Shocks (ρ)

Dataset	Years	Age-Cohort	Age-Time	Biased ML	Debiased ML
GRID	1998-2019	0.934	0.878	0.882	0.854 ± 0.075
PSID	1998-2019	0.974	0.950	0.939	0.944 ± 0.072
PSID	1978-2019	1.049	1.056	1.060	1.050 ± 0.045

Note: The table presents results on the persistence of income shocks—ρ in Section 3. This last column also presents estimates of the 95% confidence interval of the estimate from the debiased ML model. See the note for Table 1 for more detail.

For the PSID, we estimate considerably larger values for ρ. For our longer PSID sample, we estimate a value for ρ that is larger than one. This corresponds to the linear visual appearance of the (noisy) income profile for the PSID in Figure H.1. This estimate implies that income shocks are permanent. It matches the age profiles reported by Deaton and Paxson (1994) using CEX survey data, which similarly look noisy but linear.

6.2 Life-Cycle Income Inequality Across Twelve Countries

Table 4 presents estimates of the slope of the age profile of cross-sectional income variance for twelve countries using GRID data. We present estimates for the same four models as we did for the U.S. above: the conventional "age-cohort" model, the "age-time" model, and two versions of our machine learning (ML) model (with and without our debiasing procedure). As with the U.S. estimates, we estimate the age profile on the full range of ages, but then estimate its slope over the age range 35-50 years.

Consider first the estimates for the conventional age-cohort model. These vary wildly across the twelve countries. For the US, our age-cohort estimate is 0.67%. For several countries we estimate much higher slopes: for Germany the slope is 0.93%, for Italy it is 1.09%, and for Spain it is 1.67%. For other countries we estimate much lower slopes. For three countries we actually estimate negative slopes: -0.23% for Sweden, -1.37% for Brazil, and -2.16% for Argentina. Taken at face value, these results suggest enormous heterogeneity in the nature of income risk over the life-cycle across different countries.

Our estimates for the debiased ML model are much more homogeneous across countries. The

Table 4: Slope of Age Effects Across Countries

Country	Years	Age-Cohort	Age-Time	Biased ML	Debiased ML
Argentina	1996-2015	-2.16	0.80	-0.13	0.33 ± 1.95
Brazil	1993-2018	-1.37	1.33	-0.25	0.24 ± 1.63
Canada	1996-2016	0.41	0.27	0.28	0.27 ± 0.10
Denmark	1996-2016	0.50	0.08	0.28	0.30 ± 0.26
France	1996-2016	0.44	0.28	0.38	0.43 ± 0.29
Germany	2001-2016	0.93	0.26	0.67	0.73 ± 0.33
Italy	1996-2016	1.09	-0.29	0.15	0.20 ± 0.92
Mexico	2005-2019	0.13	0.82	0.20	0.43 ± 0.40
Norway	1996-2017	0.91	0.48	0.71	0.63 ± 0.19
Spain	2005-2018	1.67	0.70	0.14	0.00 ± 0.43
Sweden	1996-2016	-0.23	0.18	0.09	0.02 ± 0.12
USA	1998-2019	0.67	0.40	0.36	0.33 ± 0.13

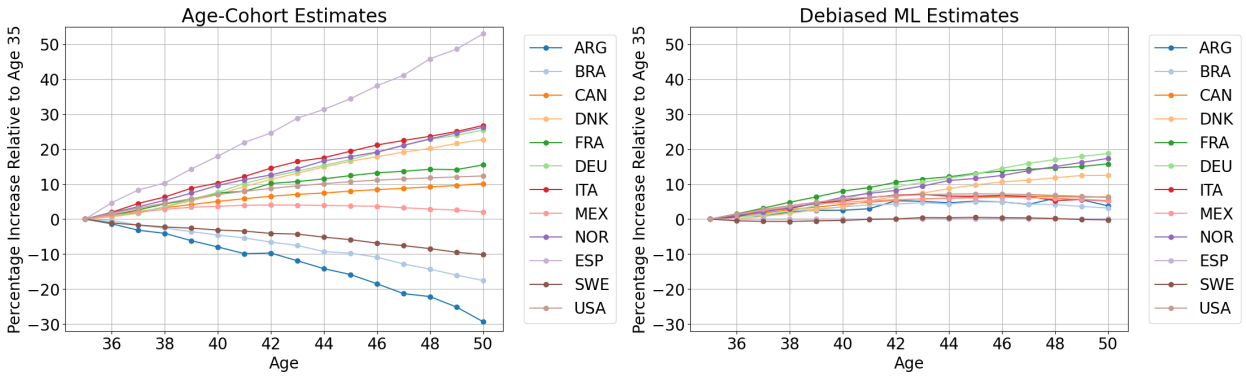
Note: The table presents results on the slope of the age effects of the cross-sectional variance of income for twelve countries using data from GRID. We present the slopes multiplied by 100 to ease readability. This last column also presents estimates of the 95% confidence interval of the estimate from the debiased ML model. See the note for Table 1 for more detail.

estimate for the US is 0.33%. The largest two slopes we estimate are 0.63% for Norway and 0.73% for Germany. None of the estimates are negative. The smallest two estimates are 0.00% for Spain and 0.02% for Sweden. Brazil and Argentina's estimates are close to the median country at 0.24% and 0.33%, respectively. This contrasts sharply with the large negative estimate for these countries for the age-cohort model.[10]

The confidence intervals we report for the debiased ML model in Table 4 are clustered by year (allowing for correlation across age within year). These confidence intervals reveal that our procedure yields quite precise estimates for a number of countries including the United States and Canada. However, for other countries, the confidence intervals we estimate are quite wide. This is particularly the case for Argentina and Brazil. For these countries, the confidence intervals include all plausible values for the slope of the age profile. These large confidence intervals arise because the proxy variables do not capture the time effects for these countries well in the earliest part of the sample, especially in 1996-1998 in Argentina and 1994-1995 in Brazil. Developing better proxy variables for time effects for these countries is an important area for further research.

Figure 6 visualizes the large difference between our estimates from the conventional age-cohort model and our debiased ML model. We plot the age profiles for all twelve countries normalized to zero at age 35 for each country. The left panel plots the estimated age profiles for the age-cohort model, while the right panel plots the age profiles for the debiased ML model. The

[10]Table H.1 presents results on the persistence of income shocks across countries.



(a) Age-Cohort Estimates

(b) Debiased ML Estimates

Figure 6: Normalized Age Profiles Across Countries

Note: The figure presents estimated age profiles for the twelve countries in GRID. The age profiles are normalized to zero at age 35 and plot the percentage changes in the cross-sectional variance of income at different ages relative to age 35.

difference is stark.

Figure 7 plots our debiased ML estimates of the age profiles for the twelve countries in levels over the full range of ages we use to estimate these profiles: ages 25 through 55 years. Several results stand out. First, there is an enormous degree of heterogeneity in the level of income variance across countries. The three Latin American countries in our dataset (Argentina, Brazil, and Mexico) have the highest income variance. Below them is the United States, followed (considerably below) by Canada and the eight European countries in our dataset.

For the three Scandinavian countries in our dataset (Denmark, Norway, and Sweden), and to a lesser extent France and Italy, we estimate a U-shaped pattern for income volatility. For the Scandinavian countries, income volatility falls substantially from age 25 to 30. Human capital models of life-cycle earnings can generate such U-shaped patterns of income volatility if workers who invest heavily in human capital trade off low initial earnings for high growth rates (Mincer, 1974; Rubinstein and Weiss, 2006).

For the other countries in our sample, the age profile of income volatility is linear or hump shaped. Interestingly, the countries with a high level of income variance have a hump shaped age profile, while the countries with smaller income variance tend to have a U-shaped profile (and the countries in the middle a linear profile). Human capital models can also generate hump shaped income profiles if the correlation between initial ability and human capital accumulation is strongly positive. Search models and learning models also imply a hump shaped age profile for income volatility (Rubinstein and Weiss, 2006).

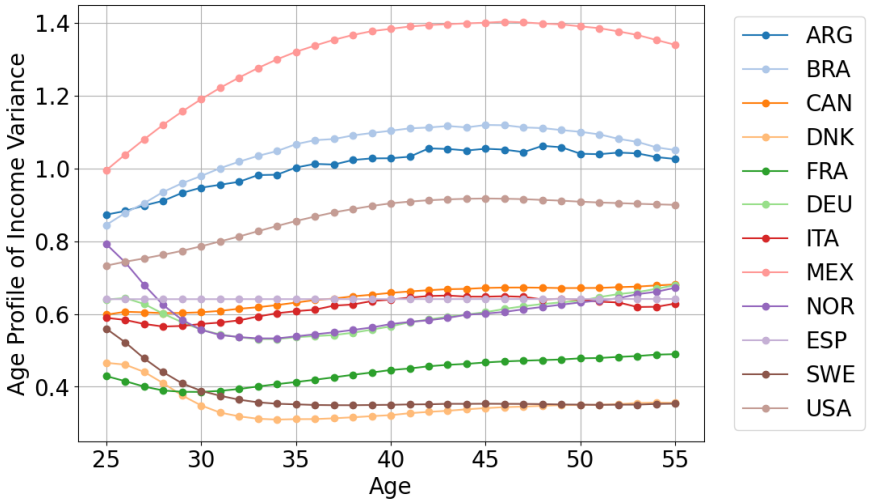


Figure 7: Age Profiles of Income Variance for an Extended Age Range

Note: The figure presents estimated age profiles for the twelve countries in GRID using our debiased machine learning procedure for the full range of ages, 25-55.

7 The Life-Cycle Profile of Average Income

We next turn to estimating the slope and shape of the life-cycle profile of average income. For the slope of the age profile of average income, we again face a severe age-time-cohort problem, just like we did for the slope of the age profile of the cross-sectional variance of income discussed above. Earlier work has made a variety of different assumptions to identify the slope of the age profile of average income (or wages). Some set time effects to zero (Guvenen et al., 2021), others set cohort effects to zero (Mincer, 1974), some report results for both of these assumptions (Huggett et al., 2011), others assume a zero slope for time effects (Deaton, 1997). Yet others adopt a simple proxy model for time effects (Gourinchas and Parker, 2002). In a study that devotes more attention than most to this issue, Lagakos et al. (2018) favor the assumption that the slope of the age profile of average wages is zero in the final years of workers' careers. This assumption dramatically flattens the age profiles of wages in some of the countries they consider (e.g., Brazil and Chile) in comparison to other identifying assumptions they report.

Both time effects and cohort effects can have important effects on the estimated slope of the age profile of average income. Rapid economic growth may result in a steeply sloping raw age profile that arises not from true age effects, but rather from time effects. On the other hand, improving education and childhood health can lead raw age profiles to appear too flat relative to their true counterparts, because younger cohorts have higher earnings. Such macroeconomic confounders are likely to be particularly important in emerging economies, with high and volatile rates of economic growth.

Our debiased proxy machine learning method provides a structured way to discriminate between the various assumptions made in the prior literature. We implement this approach in the same manner as we did for the cross-sectional variance of income earlier in the paper. The only difference is that the variable being modeled is average log income, rather than the cross-sectional variance of log income.

7.1 Life-Cycle Profile of Average Income for the United States

Table 5 presents our estimates for the age profile of average income for the same four models we consider for the cross-sectional variance of income in section 6: the "age-cohort" model (which includes age and cohort fixed effects, but no time fixed effects), the "age-time" model (which includes age and time fixed effects, but no cohort fixed effects), and two versions of our machine learning (ML) model (with and without our debiasing procedure).[11] Figure 8 plots the profile of age effects as well as the profile of year effects and cohort effects for three models: a model for which the slope of time effects is constrained to be zero (top row), a model for which the slope of cohort effects is constrained to be zero (middle row), and a model for which the slope of the age effects is constrained to be zero from age 45 to 55 (bottom row).[12]

Consider first the results in Figure 8. Notice first that the three models produce results that are rotations of each other. The age and cohort effects rotate clockwise (become less steep) as we go down the figure, while the time effects rotate counter-clockwise (become steeper). This is mechanically a consequence of the fact that $a + c = t$. A second notable result is that the profile of time effects in the top row is close to a constant. This means that setting the slope of the time effects to zero yields results that are very similar to the model where all time effects are set to zero. The same is true for cohort effects: setting the slope of the cohort effects to zero (middle row) yields results that are very similar to setting all cohort effects to zero.

A third notable result is that the model where we constrain the slope of the age effects to be zero from age 45 to 55 (bottom row), yields cohort effects that trend downward. The identifying assumption made in this bottom row is similar to the preferred identifying assumption in Lagakos et al. (2018). The bottom row of Figure 8, thus, shows that their preferred identifying assumption implies downward sloping cohort effects for the United States.

Table 5 reports that the slope of the age profile that we estimate using our debiased ML model is 3.35% per year. This is a larger slope than for any of the other models we consider. The slope

[11] The model that is selected by our model selection procedure is a kernel ridge regression with age and cohort trends, the square root of age, and the square root of the male unemployment rate.

[12] We present the age profiles for the age-cohort, age-time, and debiased ML models in Figure I.1 in the appendix.

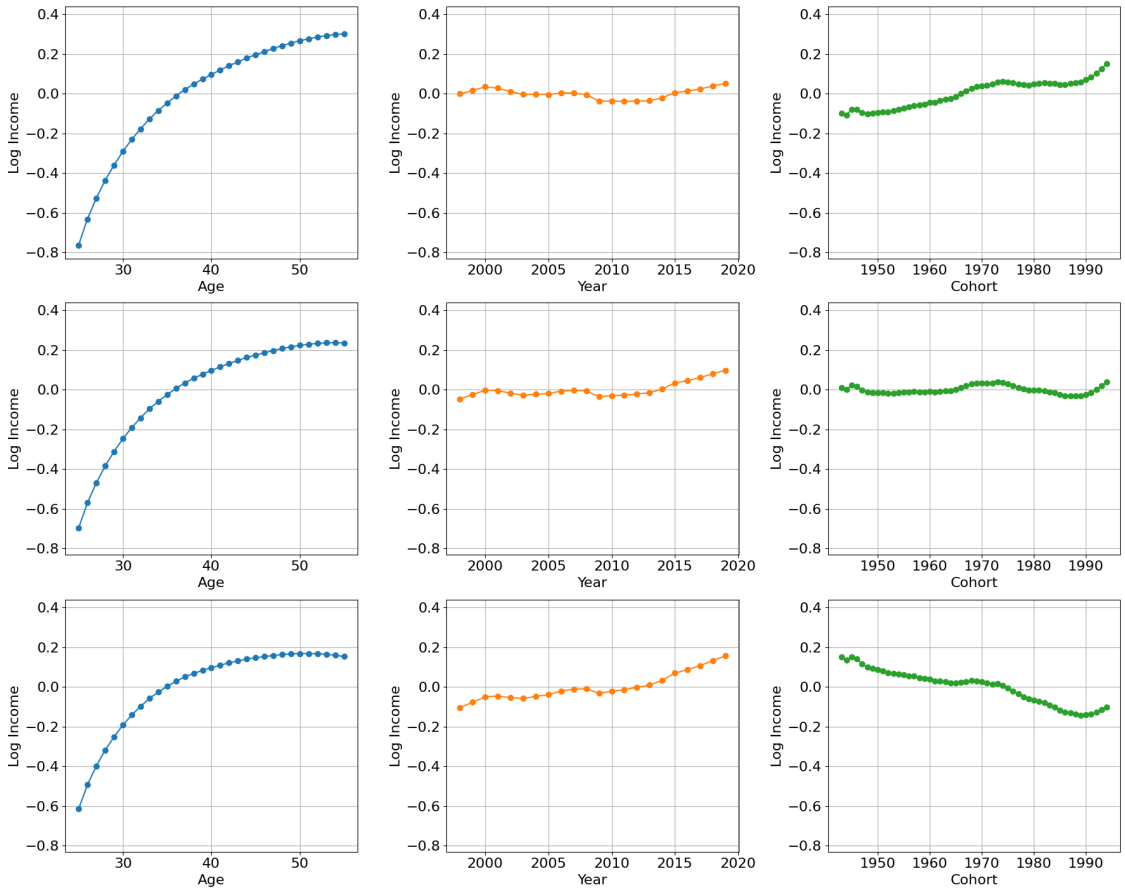


Figure 8: Age, Time, and Cohort Profiles for Average Income in United States

Note: The figure presents estimated age (left), time (center), and cohort (right) effects under three different models. In all cases, we fit a full set of age, time, and cohort dummies. In the top row, we constrain the time trend to be zero. In the middle row, we constrain the cohort trend to be zero. In the bottom row, we constrain the age effects to have zero slope from age 45 to 55.

estimated for the age-cohort model is 3.04% per year, while the slope estimated for the age-time model is 2.57% per year. The age-cohort model is very similar to the model presented in the top row of Figure 8, while the age-time model is very similar to the model presented in the middle row of Figure 8. The model presented in the bottom row of Figure 8—which constrains the slope of the age effects to be zero from age 45 to 55—yields an even smaller slope of the age profile (2.06% per year).

Among the simple models we consider, the age-cohort model provides the best approximation to our debiased ML. This contrasts with our analysis of the cross-sectional variance of income, where the age-time model provided the best approximation. Loosely speaking, one can interpret our procedure as using the proxy variables and machine learning to choose between the cohort effect and time effect estimates. For the cross-sectional variance of income, we find that time effects are more important, whereas for the average level of income, we find that cohort effects are

Table 5: Life-Cycle Profile of Average Income in the United States

Estimator	Annual Slope Age 25-55	Percentage Growth Age 25-45
Age-Cohort	3.04	97.0
Age-Time	2.57	87.8
Biased ML	2.96	93.6
Debiased ML	3.35 ± 0.44	91.9

Note: The table presents results on the slope of the age profile of average log income between ages 25 and 55 for the United States as well as the percentage change in this age profile from age 25 to 45. We present the slopes multiplied by 100 to ease readability. The "Age-Cohort" column presents results from a model with age and cohort effects, but no time effects. The "Age-Time" column presents results from a model with age and time effects, but no cohort effects. The "Biased ML" column presents results from our model without debiasing. The "Debiased ML" column presents results for our model including our debiasing procedure. For the Debiased ML model we also present estimates of the 95% confidence interval of the estimate. Results are based on data from GRID.

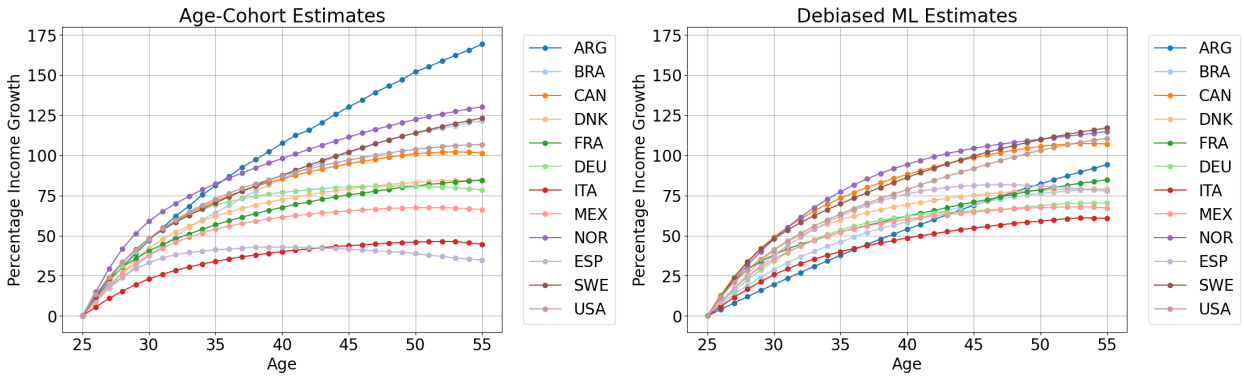
more important. Substantively, these findings suggest that business cycle shocks play a larger role driving variation in income risk and inequality over our sample than cohort factors such as early life nutrition or education, while the reverse is true for variation in average income.

Table 5 also presents results for the percentage growth of the age profile from age 25 to 45, a statistic emphasized by Lagakos et al. (2018). The difference among the four models for this statistic is not as large as for the slope of the age profile from age 25 to 55. The reason for this is that much of the difference in the curvature of the age profiles we estimate arises after age 45.

7.2 Life-Cycle Profile of Average Income for Twelve Countries

Table 6 and Figure 9 present estimates of the age profile of average income for 12 countries. Figure 9 shows graphically that our debiased ML model generates much less dispersion in age profiles for average income across these 12 countries than the conventional age-cohort model. The percentage growth from age 25 to 55 ranges from 61% to 117% for the debiased ML model, while it ranges from 35% to 169% for the age-cohort model. In this regard, our results for average income are similar to our results for the cross-sectional variance of income earlier in the paper.

Turning to Table 6, we see that there is often a substantial difference between the slope estimated for the age-cohort model and the age-time model. While the age-cohort model yields a larger slope for most countries (the average age-cohort slope is 2.77% per year while the age-time slope is 2.09% per year), the age-time model yields a substantially larger slope for Spain and a larger slope for Germany and Italy as well. For most countries, our debiased ML model yields a slope that is closer to the age-cohort model than the age-time model. But this is not the case for all countries. In particular, the debiased ML model yields results that are closer to the age-time



(a) Age-Cohort Estimates

(b) Debiased ML Estimates

Figure 9: Age Profiles in Levels Across Countries

Note: The figure presents estimated age profiles of income (in levels) for the twelve countries in GRID. The profiles are offset to start at 0 and the y-axis is multiplied by 100 so the profile can be read as percentage growth.

model for Argentina, Brazil, and Spain. The slope estimates from the age-cohort model are quite high for Argentina and Brazil. A natural explanation for this is that time effects are important in emerging markets.

For most countries, the 95% confidence interval of our estimate for the debiased ML model is of a similar size to that for the United States. However, for Argentina and Brazil, the 95% confidence intervals are much larger. Particularly for Brazil, the confidence interval is very large implying that we cannot rule out any plausible values for the slope of the age profile. This large confidence interval for Brazil arises due to the inclusion of the earliest years of the sample when Brazil was experiencing a hyperinflation. Our proxy model does a poor job at explaining the variation in the time effects during this very volatile period. As a result, the model generates large errors in the out-of-sample predictions for this period and we estimate a large confidence interval.

7.3 Is the Age Profile Steeper in High-Income Countries?

The central question Lagakos et al. (2018) investigate is whether "experience effects"—the income gain from an additional year of work experience (which they proxy by age)—are larger in high-income countries than in low-income countries. Whether this is the case sheds valuable light on the predictions of several different classes of models that seek to explain income differences across countries. For example, Manuelli and Seshadri (2014) build a model in which higher productivity raises the returns to human-capital investment. This naturally generates the prediction that richer countries have steeper income profiles.

The results in Lagakos et al. (2018) rely on the identifying assumption that the slope of the age

Table 6: Slope of Age Profile of Average Earnings for 12 Countries

Country	Years	Age-Cohort	Age-Time	Biased ML	Debiased ML
Argentina	1996-2015	5.35	1.88	3.29	3.14 ± 1.03
Brazil	1993-2018	3.89	2.14	1.80	2.70 ± 3.74
Canada	1996-2016	2.90	2.18	3.05	3.09 ± 0.13
Denmark	1996-2016	2.37	1.99	2.11	2.19 ± 0.41
France	1996-2016	2.29	1.92	1.88	2.30 ± 0.46
Germany	2001-2016	2.03	2.33	1.84	1.91 ± 0.33
Italy	1996-2016	1.81	2.55	1.50	1.81 ± 0.42
Mexico	2005-2019	1.73	1.39	1.69	1.83 ± 0.38
Norway	1996-2017	3.62	1.89	3.46	3.35 ± 0.42
Spain	2005-2018	0.60	2.49	2.10	2.33 ± 0.45
Sweden	1996-2016	3.61	1.71	3.26	3.40 ± 0.62
USA	1998-2019	3.04	2.57	2.96	3.35 ± 0.44

Note: The table presents results on the slope of the age effects of the cross-sectional mean of log income for twelve countries using data from GRID. We present the slopes multiplied by 100 to ease readability. This last column also presents estimates of the 95% confidence interval of the estimate from the debiased ML model. See the note to Table 5 for more detail.

profile of average income (wages in their case) is zero at the end of the working life. They motivate this assumption on theoretical grounds. But it is not clear that this is a robust implication of theory. Learning and human capital accumulation can continue as a byproduct of work even if workers' incentive to learn fall to zero. Many firms have a job ladder with their most experienced workers occupying the top rungs of the ladder, having the most responsibility, and being paid the most. In fact, our results in section 7.1 show that Lagakos et al.'s identifying assumption is clearly rejected for the United States.

We revisit the relationship between the slope of the age profile of average earnings and GDP without assuming that the slope of the age profile is zero at the end of working life. Figure 10 presents a scatter plot of the relationship between a country's log GDP per capita measured at PPP in 2015 and the 20-year percentage growth in average income between age 25 and 45 estimated using our debiased machine learning procedure. This figure is analogous to Figure 3 in Lagakos et al. (2018). The figure demonstrates a clear positive relationship: higher income countries have steeper age profiles for average income. We therefore confirm the key finding of Lagakos et al. (2018) with entirely different data even though we find evidence against their identifying assumption.

Table 7 presents estimates of the slope of the relationship between the 20-year percentage growth in average income from age 25 to 45 for different countries and the country's GDP per capita measured at PPP. Each row reports the slope of the line through the equivalent of Figure 10,

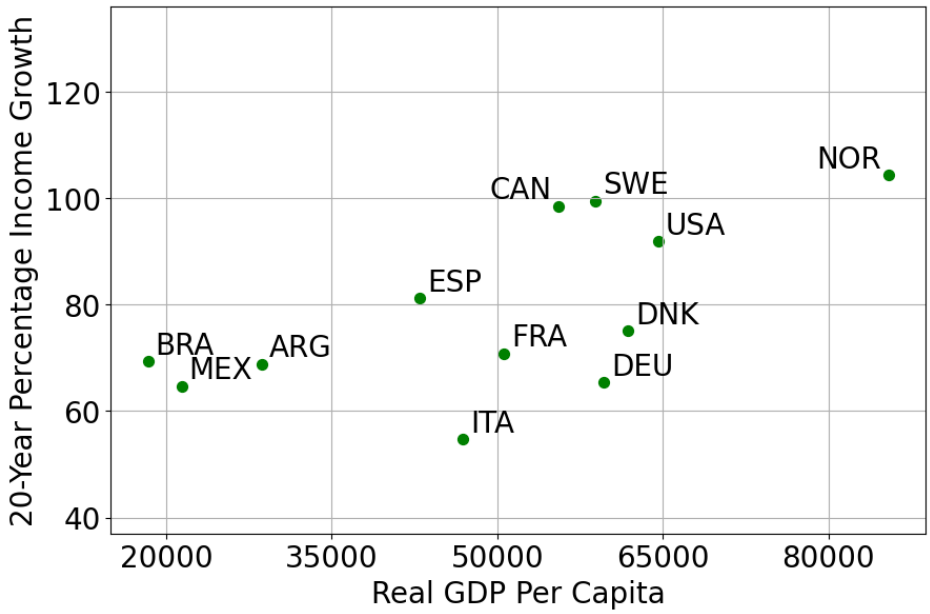


Figure 10: Income Growth vs GDP Per Capita

Note: The figure plots the 20-year percentage growth in average income from age 25 to 45 (100 times the log difference) for different countries in our sample estimated using our debiased machine learning model against real GDP per capita measured at PPP in 2015.

but varying how the age profile is estimated. For our debiased ML model, this coefficient (estimated using OLS) is 23.6. This value is quite similar to the analogous coefficient of 26.2 reported in Lagakos et al. (2018) for their Figure 3. Our result, thus, not only confirms their result qualitatively, but is actually also quantitatively quite similar. Table 7 also presents estimates of this regression coefficient for other models. The relationship is positive for all models we consider but largest for our debiased ML model. In other words, applying what we argue is a more plausible identifying assumption to address the age-time-cohort problem actually yields stronger evidence in favor of the relationship between experience effects and economic development.

8 Conclusion

We propose a new proxy variable machine learning approach to disentangle age, time and cohort effects. We apply this methodology to two classic problems: the estimation of the age profile of cross-sectional variance of income and of average income over the lifecycle. Our benchmark empirical analysis applies our methodology to the newly developed Global Repository of Income Dynamics (GRID) data, a collection of aggregate statistics constructed using administrative microdata for 12 countries: Argentina, Brazil, Canada, Denmark, France, Germany, Italy, Mexico, Norway, Spain, Sweden and the United States.

Table 7: Income Growth vs GDP Per Capita

	OLS Coefficient	R^2
Age	19.7	0.527
Age-Cohort	13.1	0.033
Age-Time	19.1	0.540
Flat Last 10	19.2	0.254
Debiased ML	23.6	0.292

Note: The table presents estimates of the coefficient in an OLS regression of 20-year percentage growth in average income between ages 25 and 45 for different countries on the country's log GDP per capita at PPP. We present this regression coefficient for a model with only age fixed effects (Age), age and cohort fixed effects (Age-Cohort), age and time fixed effects (Age-Time), age, time, and cohort effects in which the slope of the age effects are set to zero over ages 45 to 55 (Flat Last 10), and our debiased ML model.

Our methodology allows us to make substantially weaker identifying assumptions than standard approaches in the literature, and it yields interesting new results about household income dynamics. For the cross-sectional variance of income, we find that income inequality within cohorts grows at a rate that is roughly half as large as standard methods imply. Switching from PSID data to GRID data also leads to a substantial reduction in the slope of the age profile of income inequality. Moreover, our estimates are much less erratic across countries and consistent with economic theory than those derived from conventional methods. These findings imply that time effects—which are ruled out by conventional methods—are important for understanding the evolution of the cross-sectional variance of income.

For the average level of income, we estimate a steeper age profile for the United States than conventional methods. Looking across countries, we estimate that high-income countries have steeper age profiles than low income countries. Our estimate of this relationship is qualitatively and quantitatively in line with the results of Lagakos et al. (2018). Interestingly, we arrive at this result even though we find evidence against the identifying assumption made in Lagakos et al. (2018).

Our applications demonstrate a number of methodological points relevant for other applications. First, in panel data settings with persistent macroeconomic data, it is crucial to use "blocked" cross-validation as opposed to off-the-shelf cross-fitting techniques from the machine-learning literature that simply leave out a random fraction of the data. Second, when choosing a proxy model for time effects, we must restrict attention to proxy variables that differ sufficiently from time trends. Finally, we find that kernel ridge regression performs well in our application (with our relatively modestly-sized dataset) relative to more complex machine learning models such as random forests or neural networks. Our methodology aims to offer a disciplined alternative to existing methods in the social sciences for addressing the age-time-cohort problem in a

diverse set of applications ranging from the age profile of political preferences to fertility or firm innovation.

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A Data

A.1 GRID US

For the results reported in Section 6.1, we use data for the United States released as part of the Global Repository of Income Dynamics (GRID) (Guvenen et al., 2022a). The data come from the Longitudinal Employer-Household Dynamics Infrastructure files at the Census Bureau. The GRID dataset consists of certain aggregates calculated using this microdata. We do not have access to the underlying microdata, only the aggregates reported in the GRID data. These aggregates are calculated using observations at the individual level. We consider only aggregates for males for our main specification. The income measure used is log labor earnings including wages, salaries, tips, bonuses, but not self-employed or business income deflated by the PCE price deflator. Observations are kept if they are above a floor of 260 times the real federal hourly minimum wage in that year. Income is winsorized at the 99.999999th quantile. See Guvenen et al. (2022b) and the GRID documentation for more details.

A.2 PSID following GRID

In section Section 6.1, we also present results based on data from the PSID. We develop sample selection criteria intended to mimic those used by the GRID dataset discussed above. We use the individual file to track all male individuals in the PSID over time, regardless of their head of household or marital status. We consider only individuals from the representative SRC sample (i.e., we drop the poverty and immigrant oversamples). We use log labor income deflated by the PCE price deflator as the income measure. We keep only observations above 260 times the real federal hourly minimum wage. We winsorize observations above the 99.99 quantile. Finally, log income is residualized by a regression on year and age dummies.

A.3 GRID International

In section Section 6.2, we present results for 11 additional countries available in the GRID database. GRID tries as much as possible to harmonize the data across countries, but the precise sample selection criteria and variable definitions differ slightly from country to country. The GRID website provides details on sample selection and variable definitions for each country — https://www.grid-database.org/documentation. We provide three illustrative examples here that highlight some of the differences across countries.

The data for Germany combine records from social security data and personal income tax records. GRID uses a random 10% subsample of social security observations, and a random 25% subsample of tax observations. Annual earnings includes overtime pay, bonuses, 13th month pay, paid sick leave, severance pay, and vacation allowance. Self-employment is excluded. A wage floor is imposed based on the Germany national minimum wage.

By contrast, the data for Norway come from annual tax records, which cover the entire population. Labor income includes salaries and hourly wages, fees received by board members, bonuses, commissions, overtime, piecework, performance, caregiver, severance, and holiday payments, fixed wage and irregular supplements. Self-employment is excluded. Norway does not have a minimum wage, so a wage floor is imposed based on the US minimum wage.

Finally, the data for Argentina come from employer-employee matched social security records. GRID uses a random 3% subsample, which is representative of formal employment at private firms — however private formal employment is only 30-40% of total employment. Labor income includes base salary, overtime compensation, performance and seasonal bonuses, paid vacations, paid sick leaves, and severance payments. Self-employment is excluded. A wage floor is imposed based on the Argentinian national minimum wage.

The sample period available for different countries in GRID also varies somewhat. Table A.1 provides the sample period for each country.

Table A.1: GRID Sample Years

Country	Full Sample Years	Sample Years with Proxies
Argentina	1996-2015	1996-2015
Brazil	1985-2018	1993-2018
Canada	1983-2016	1996-2016
Denmark	1987-2016	1996-2016
France	1991-2016	1996-2016
Germany	2001-2016	2001-2016
Italy	1985-2016	1996-2016
Mexico	2005-2019	2005-2019
Norway	1993-2017	1996-2017
Spain	2005-2018	2005-2018
Sweden	1985-2016	1996-2016
USA	1998-2019	1998-2019

A.4 Proxy Variables

Our baseline set of proxy variables are ten macroeconomic time series: log GDP, log consumption, log investment, export values, log industrial production, male unemployment, log CPI, a short term interest rate, the exchange rate, and oil prices. We have sought to obtain harmonized definitions for these time series across countries. We source GDP, consumption, investment, exports, unemployment, and the CPI from the World Bank's World Development Indicators (WDI) database. GDP, consumption, investment are measured in constant 2015 US dollars. For exports, we use the export value index (2015 = 100). For unemployment, we use the percent unemployed out of the male labor force because our income data is restricted to men. We source data on industrial production from the International Monetary Fund's (IMF) International Financial Statistics (IFS) database. Industrial production data are not available for Argentina and Denmark. For the short-term interest rate, we choose one of the following three interest rates for each country: the T-bill rate, the money market rate, and the policy rate. For each country, we choose the rate that has the longest sample. We source the T-bill and money market rates from Bloomberg, and the policy rate from the IMF IFS database. For the exchange rate, we use the real effective exchange rate series from Darvas (2021). We use their broad measure, REER 170. Finally, we use the same oil price series across countries. We adopt the West Texas Intermediate (WTI) spot crude oil price from FRED, Federal Reserve Bank of St. Louis.

For the United States, we include a measure of the skill premium. From the Current Population Survey, we source median usual weekly earnings of full-time wage and salary workers age 25 years and over with a Bachelor's degree and higher, and with a high school diploma, but no college. We use the ratio of these two series as our measure of the skill premium.

B Proxy Assumptions

The original idea of proxying for time effects goes back to at least Farkas (1977) who replaced calendar year with the unemployment rate but without giving any formal justification. Heckman and Robb (1985) provides a formal justification from a structural perspective, and introduces the term "proxy variable." This framework was substantially extended by Winship and Harding (2008), who provided a new formal justification based on Pearl's "front door criterion" (Pearl, 2009): as long as we observe *all* of the relevant mediators for at least one of time or cohort, then the age profile is identified.

Figure B.1 presents a diagram of the causal structure that we assume following Heckman and

Robb (1985). Notice that we do not do any modeling of the mechanisms that sit between age and the outcome, or between cohort and the outcome. Recall, this is without loss of generality, because we only need to cut-off a single one of A, T, or C from Y in order to identify the age profile. We choose T because we believe the macroeconomic mediators for T are easier to identify than those for A or C in our setting. Importantly, this structure allows a hypothetical mechanism like education to affect Y. We just assume that all such mechanisms are captured by a combination of age and cohort effects.

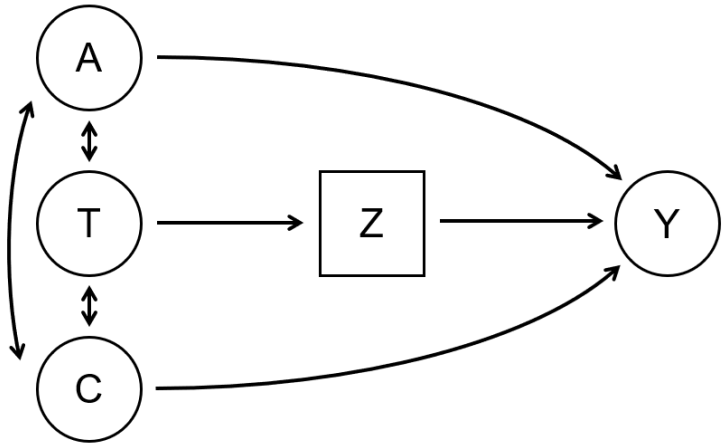


Figure B.1: Proxy Variable Structure

Ultimately, this causal diagram boils down to two main substantive assumptions. First, an exclusion restriction: we assume that, holding A and C fixed, T has no effect on Y except through Z. For example, this means there are no additional unobserved time-varying macroeconomic variables that cause Y. Second, we assume that neither A nor C effect Z. This prevents scenarios where Z itself has an age-time-cohort problem. Note that we do *not* need to rule out, for example, an age composition effect on the unemployment rate Z. Time-varying age composition would itself be a time effect in our framework.

We could relax the second assumption above using techniques from Winship and Harding (2008). For example, we could assume that the proxies Z can be affected by *at most* two of A, T, and C. This would introduce additional complexity in the form of a first stage regression, as discussed in Winship and Harding (2008). As a consequence, we instead make the stronger assumption embedded in Figure B.1. Extending our machine learning estimator to work for any causal diagram compatible with Winship and Harding (2008) is one possible direction for future work. If any of the proxies Z are affected by all three (A, C, T) the model is no longer identified. In this case, we would need to find intermediate proxies that are plausibly affected by only two of

the three.

Finally, formal justification via the front-door criterion requires stochasticity of the mediators Z.[13] In other words, the proxies Z cannot be perfectly deterministic functions of T. Otherwise we'd be back to some form of multicollinearity. This connects directly to our derivation of the Riesz representer in Appendix C.3: if the proxies are deterministic functions of T, then our theoretical analysis indicates the slope of the age profile is unidentified. This observation motivates our requirement that any proxy variables must have an R^2 of less than 0.75 in the regression of T on Z.

While the approach we adopt relies on non-trivial assumptions, these assumptions are much weaker than the assumptions made in most existing work that seeks to solve the age-time-cohort problem in economics. It will be interesting to see if these assumptions can be further relaxed or empirically validated in future work.

C Intuition for Riesz Representers and Double Machine Learning

C.1 Double/Debiased Machine Learning

The machine learning prediction task (by minimizing the mean squared error) tries to estimate the conditional expectation:

$$m_0(a, c, z) \coloneqq \mathbb{E}[\sigma^2_{at}|a, c, z].$$

Typically, our final machine learning model $\hat{m}(a, c, z)$ introduces bias to reduce the variance of the estimator. Mechanically, this means that we do not make the in-sample prediction residuals as small as we possibly could, given a highly flexible machine learning function class. This is usually necessary to control overfitting and successfully generalize out of sample.

The fundamental idea behind double/debiased machine learning is to exploit the fact that our actual estimand of interest is the slope, slope(m_0)—a single scalar—which is much simpler than the whole function $m_0(a, c, z)$. We do not need a completely unbiased estimate of the entire function m_0 (which would have prohibitively large variance). We only need an unbiased estimate of slope(m_0). This should be easier to achieve.

The bias of machine learning estimators results from these methods not making the prediction residuals as small as possible. For example, while OLS produces the smallest possible residuals among linear models, Lasso and Ridge are regularized, creating bias toward a simpler model and

[13] We thank Lihua Lei for pointing this out.

resulting in larger residuals. Double/debiased machine learning cleverly addresses this regularization bias by correcting the prediction residuals *but only if they matter for the slope*. This works by adding a bias correction term to the initial estimate of the slope, slope$(\hat{m})$:

$$\text{slope}(\hat{m}) + \mathbb{E}\Big[\nabla\text{slope}(a,c,z) \cdot (\underbrace{\sigma^2_{at} - \hat{m}(a,c,z)}_{\text{prediction errors}})\Big]. \tag{10}$$

What is ∇slope(a,c,z)? It is the gradient of the slope when viewed as an operator, slope$(\cdot)$, that takes a *function* of a,c,z such as $\hat{m}$ as its input. Note that because the input to slope$(\cdot)$ is a function of a,c,z, the gradient, ∇slope, is also a function of a,c,z. We will define this gradient more formally in the next section. For now think of the gradient intuitively: it measures how much changing $\hat{m}(a,c,z)$ at a particular value of a,c,z impacts the final estimate slope$(\hat{m})$. For example, we would expect that the observations where a is at the extremes of the age range would be more impactful for the slope than observations in the middle of the age range. Using this interpretation, when we do debiasing in Equation (10), we correct the residuals more if the prediction at that point is very influential for the final slope estimate.

To understand why (10) corrects bias, notice that it takes the form of a first-order Taylor expansion. The familiar Taylor approximation from elementary calculus is: $f(x) \approx f(a) + f'(a)(x-a)$. In expression (10), slope$(\cdot)$ corresponds to $f(\cdot)$, and $\hat{m}$ corresponds to a. But what about x? Here we can apply the law of iterated expectations: $\mathbb{E}[\sigma^2_{at}] = \mathbb{E}[\mathbb{E}[\sigma^2_{at}|a,c,z]] = \mathbb{E}[m_0(a,c,z)]$. When we plug this expression back into (10), we get:

$$\text{slope}(\hat{m}) + \mathbb{E}\Big[\nabla\text{slope}(a,c,z) \cdot (m_0(a,c,z) - \hat{m}(a,c,z))\Big].$$

So the debiased estimate is a first-order Taylor approximation of the true population estimand slope(m_0), where the Taylor approximation is taken around our initial, biased machine learning estimate, slope$(\hat{m})$. Furthermore, we can compute this bias correction term even though we do not know the true m_0 in advance. The following is a simple way to think about the end-to-end process:

1. We want to correct the residuals but only if they matter for estimating the slope.
2. One way to do this is to multiply the residuals by the "gradient of the slope with respect to the observation".
3. We end up with a Taylor series approximation for the true estimand, slope(m_0).

See Kennedy (2022) for a more detailed theoretical review.

C.2 "Riesz Representers" Are Gradients with Respect to Function-Valued Inputs.

In the previous section, we describe a Taylor approximation using $\nabla\text{slope}(a, c, z)$. This is the gradient of the slope operator, $\text{slope}(\cdot)$, with respect to its input, a function of a, c, z. The crucial observation is that the slope is a *linear* functional. For any two functions $m_1(a, c, z)$ and $m_2(a, c, z)$, and any $a, b \in \mathbb{R}$,

$$\text{slope}(am_1 + bm_2) = a \cdot \text{slope}(m_1) + b \cdot \text{slope}(m_2).$$

Linear functionals have a special property. Just like a linear function defined on $\mathbb{R}^d$ can be written in terms of coefficients in $\mathbb{R}^d$, a linear functional defined over functions $m(a, c, z)$ can be written in terms of "coefficients" that are themselves a function of a, c, z. This isn't true for completely arbitrary spaces of functions. In particular, we need the following restriction on the functions m that we consider: $\mathbb{E}[m(a, c, z)^2] < \infty$. We denote the set of functions that satisfy this condition L_2. Functions in L_2 have an inner product just like vectors in $\mathbb{R}^d$ have an inner product. This inner product has the following form. For any $f, g \in L_2$:

$$\langle f, g \rangle \coloneqq \mathbb{E}[f(a, c, z)g(a, c, z)].$$

Because the slope is a linear functional, there exist unique "coefficients" in L_2 that we will denote $\alpha_0(a, c, z)$ such that:

$$\langle \alpha_0, m \rangle = \text{slope}(m), \forall m \in L_2. \tag{11}$$

In $\mathbb{R}^d$, the gradient of a linear function is equal to the coefficients in $\mathbb{R}^d$. Similarly, the gradient $\nabla\text{slope}(a, c, z) = \alpha_0(a, c, z)$. Spaces of general functions that have an inner product are called "Hilbert spaces" and the "coefficients" in these spaces are called the "Riesz representer". For a review of Riesz representers for an applied audience in epidemiology see Williams et al. (2025).

C.3 Riesz Representer for the Slope Operator

So far, we have reviewed how to debias a machine learning estimate using the Riesz representer. However, we have discussed the Riesz representer abstractly. In this section, we present the Riesz representer for our particular object of interest: the slope of the age profile. The Riesz representer for the slope of the age profile has an explicit analytic form. In this section, we present the analytic

form but defer the derivation.

Define $\vec{a}$ to indicate the vector of ages $a_{\min}, ..., a_{\max}$. The Riesz representer of the slope is:

$$\alpha_{\text{slope}}(a, c, z) = \underbrace{\frac{a - \mathbb{E}[A]}{(\vec{a} - \mathbb{E}[A])^\top (\vec{a} - \mathbb{E}[A])}}_{\text{age slope weights}} \underbrace{\frac{1}{P(T = c + a | Z = z)}}_{\text{inverse time propensity}}. \tag{12}$$

There are two contributions to the size of α_{slope}: the age slope weights and the inverse time propensity $1/P(T = c + a|Z = z)$. We will interpret these in turn.

The age slope weights correspond precisely to running OLS of the age profile against age. Define the vector $\tilde{\mathbf{a}} = \vec{a} - \mathbb{E}[A]$. Then for any vector $\mathbf{y}$ of the same length, the OLS coefficient for the regression of $\mathbf{y}$ on age with an intercept is:

$$\frac{\tilde{\mathbf{a}}^\top \mathbf{y}}{\tilde{\mathbf{a}}^\top \tilde{\mathbf{a}}}.$$

The term $\tilde{\mathbf{a}}/\tilde{\mathbf{a}}^\top \tilde{\mathbf{a}}$ is precisely the vector of age slope weights in (12) for each a.

The entries of these weights take on the largest magnitude values at the earliest and latest ages. Recall that when $\alpha_{\text{slope}}(a, c, z)$ is large for a given observation, then that observation is especially impactful for estimating the slope. Intuitively, the age slope weights reflect the fact that observations with ages farthest from the mean age have the most impact on the estimated slope.

The second term in (12) is the inverse time propensity. This term helps demonstrate precisely how proxies achieve identification. If $P(T = c + a|Z = z)$ is not bounded away from 0, the Riesz representer α_{slope} will be unbounded. Using the gradient intuition, if the Riesz representer is unbounded, it means that an arbitrarily small change at that data point can cause an arbitrarily large change to the slope estimate. In other words, the age slope is identified if and only if $P(T = c + a|Z = z)$ is not 0.[14]

Consider using a time trend as the proxy Z, i.e. $Z = T$. In this case there is no uncertainty about time conditional on the proxy; $P(T = c + a|Z = z)$ either equals 0 or 1. Therefore the denominator is not always greater than 0 and the slope is not identified.

Proxy variables can only achieve identification by introducing some uncertainty into the relationship between Z and T. First, notice that this motivates our use of blocked cross-validation. Consider trying to estimate $P(T = c + a|Z = z)$ from the data. Without sample splitting, for continuous valued proxies there is usually a bijective map between Z and T. E.g., consider taking

[14] More formally, the requirement is that α_{slope} is square integrable, which requires the inverse time propensity term not to blow up.

Z to be inflation. Inflation in the US in the year 2005 was 3.39275. There is no other year in which inflation takes on *precisely* this value, so without any sort of sample splitting, we could construct a map that predicts T given inflation with perfect accuracy. $P(T = c + a|Z = z)$ would again always be 0 or 1, and the slope would be unidentified. Instead, if we always split the observations into train and test folds by years, then it is generally not possible to perfectly predict T given Z out-of-sample. As a result $P(T = c + a|Z = z)$ will be greater than 0, and the slope becomes identified.

Second, notice that this means that proxies that are *approximately* but not perfectly trending, such as GDP, yield very poor identification. When Z is GDP, then $P(T = c + a|Z = z)$ is always close to 0 or 1, even when sample splitting by year. All identification comes from the deviations around the trend, and these observations correspond to large values of the Riesz representer. In other words, the estimate of the slope is driven by these deviations from the trend. During debiasing, the Riesz representer is applied as a set of weights, and reweighting by large values translate into large standard errors — correctly reflecting poor identification. For time series like unemployment or inflation, it is generally harder to predict T as a function of Z, resulting in a $P(T = c + a|Z = z)$ that is strictly bounded away from 0. In this case the Riesz representer does not take on extremely large values, and we achieve better identification.

Therefore, the discussion above based on (12) motivates our choices to exclude proxy variables that are very strongly trending (with an $R^2 > 0.75$ in the regression of T on that proxy), and to perform cross-validation by splitting whole years into train and test folds.

Finally, note that instead of minimizing the Riesz loss as described in the previous section, we could instead estimate the conditional density $P(T|Z)$ and then plug in our estimate into (12) to compute the Riesz representer for debiasing. This has two problems. First, conditional density estimation becomes statistically very challenging as the dimensionality of Z grows. Second, taking the inverse in (12), $s(a)/\hat{P}(T = c+a|Z = z)$, has undesirable properties. Even if $\hat{P}(T = c+a|Z = z)$ can be estimated well, we could end up with a poor estimate of $1/\hat{P}(T = c + a|Z = z)$ due to instability for small values of $\hat{P}(T = c+a|Z = z)$. For these reasons, we prefer to directly estimate the Riesz representer using the Riesz loss. However, (12) provides important intuition about the underlying structure of the identification problem.

C.4 Estimating the Riesz Representer

Finally, we comment on how we estimate $\hat{\alpha}$. We discussed above how α_0 for the slope functional has a closed-form expression. Note that in the expression (12), the only unknown quantity is the

conditional probability in the denominator — $P(T|Z)$. One simple way to estimate α_0, would be to first get an estimate of this probability, $\hat{P}(T|Z)$, and then to compute the "plug-in" estimate:

$$\hat{\alpha}(a,c,z) = \frac{a - \mathbb{E}[A]}{(\vec{a} - \mathbb{E}[A])^\top (\vec{a} - \mathbb{E}[A])} \frac{1}{\hat{P}(T = c + a | Z = z)}.$$

This strategy is called "inverse probability weighting" and is used in Chernozhukov et al. (2018).

However, the recent literature on debiased machine learning has emphasized that this is usually *not* a reliable way to estimate the Riesz representer—see for example Ben-Michael et al. (2021) and Chernozhukov et al. (2022). Among other issues, inverting the estimated probabilities means that when $P(T|Z)$ is close to zero, very small estimation errors can lead to enormous estimation errors in $\hat{\alpha}$.

Instead in this paper, we adopt the "Riesz loss" from Chernozhukov et al. (2022). The true Riesz representer α_0 is the *unique* minimizer of the optimization problem:

$$\min_{\alpha \in L_2} \{\mathbb{E}[\alpha(a,c,z)^2 - 2 \cdot \text{slope}(\alpha)]\}. \tag{13}$$

Minimizing this loss is equivalent to minimizing the squared-loss between α and α_0:

$$\begin{aligned}
&\min_{\alpha \in L_2} \mathbb{E}[(\alpha(a,c,z) - \alpha_0(a,c,z))^2] \\
&\quad = \min_{\alpha \in L_2} \mathbb{E}[\alpha(a,c,z)^2 - 2 \cdot \alpha(a,c,z)\alpha_0(a,c,z) + \alpha_0(a,c,z)^2] \\
&\quad = \min_{\alpha \in L_2} \mathbb{E}[\alpha(a,c,z)^2] - 2 \cdot \mathbb{E}[\alpha(a,c,z)\alpha_0(a,c,z)] \\
&\quad = \min_{\alpha \in L_2} \mathbb{E}[\alpha(a,c,z)^2] - 2 \cdot \langle \alpha, \alpha_0 \rangle \\
&\quad = \min_{\alpha \in L_2} \mathbb{E}[\alpha(a,c,z)^2 - 2 \cdot \text{slope}(\alpha)],
\end{aligned}$$

where we applied the key property of the Riesz representer (11) for:

$$\text{slope}(\alpha) = \langle \alpha, \alpha_0 \rangle, \forall \alpha \in L_2.$$

Finally, we connect this optimization problem back to the original intuition, where we referred to the Riesz representer as the gradient $\nabla \text{slope}(a,c,z)$. If we take the first order condition of the Riesz loss optimization problem (13), we get that the solution α^* must satisfy:

$$2\alpha^*(a,c,z) - 2\nabla \text{slope}(a,c,z) = 0 \implies \alpha^*(a,c,z) = \nabla \text{slope}(a,c,z).$$

C.5 Formal derivation of the age slope Riesz representer

For completeness, we include a formal derivation of the closed-form for the Riesz representer in (12). We will make the simplifying assumption that cohort c appears in every year t in the data. We show how to relax that assumption at the bottom of the derivation at the cost of extra complexity.

For a given $m(a, c, z)$, the slope of the age profile (using the standard OLS math with an intercept) is:

$$\begin{aligned}\text{slope}(m) &= \sum_a \frac{a - E[A]}{(\vec{a} - E[A])^\top (\vec{a} - E[A])} E[m(a, C, Z)] \\ &= E\left[\left(\frac{A - E[A]}{(A - E[A])^\top (A - E[A])}\right) \frac{1}{P(A|C, Z)} m(A, C, Z)\right]\end{aligned}$$

where $E[m(a, C, Z)]$ is the value of the age profile at age a as we defined in Section 5.3. The changing of $m(a, C, Z)$ to $m(A, C, Z)$ in the second line uses the standard inverse propensity score argument.

Thus we have from the definition of the Riesz representer:

$$\alpha_{\text{slope}}(a, c, z) = \frac{a - E[A]}{(\vec{a} - E[A])^\top (\vec{a} - E[A])} \frac{1}{P(A = a|C = c, Z = z)}.$$

Next conditional on $C = c$, the events $A = a$ and $T = a + c$ coincide. Therefore we have:

$$P(A = a|C = c, Z = z) = P(T = a + c|C = c, Z = z).$$

Finally, if cohort c is represented in every sampled year, we have that $P(T|C, Z) = P(T|Z)$. This completes the derivation of (12).

If instead, cohort c only appears in a subset of total years, $\mathcal{T}_c$, then conditioning on C changes the probability. Applying Bayes' rule we get:

$$P(T = a + c|C = c, Z = z) = \frac{P(T = a + c|Z = z)}{P(T \in \mathcal{T}_c|Z = z)},$$

so there is an additional adjustment term to the inverse time propensity in (12) depending on $\mathcal{T}_c$.

C.6 Semiparametric Efficiency Bound

One useful consequence of having derived the closed-form of the Riesz representer is that we can use it to characterize the *semiparametric efficiency bound* — a lower bound on the asymptotic

variance of *any* regular estimator of the target parameter. The technique is standard — see e.g. Chernozhukov et al. (2023). We include the bound here for completeness.

Define:

$$s(a) \coloneqq \frac{a - E[A]}{(a - E[A])^\top (a - E[A])},$$

the OLS weights from the previous section. Then the semiparametric efficiency bound is:

$$V_{\text{eff}} = \mathbb{E}\left[\left(\sum_a s(a) m_0(a, C, Z) - \text{slope}(m_0) + \alpha_{\text{slope}}(A, C, Z)(\sigma^2_{AT} - m_0(A, C, Z))\right)^2\right].$$

Plugging in our definition of α_{slope} and simplifying we get:

$$V_{\text{eff}} = \text{Var}\left(\sum_a s(a) m_0(a, C, Z)\right) + \mathbb{E}\left[\sum_a \frac{s(a)^2 \text{Var}(\sigma^2_{AT} | A = a, C, Z)}{P(T = a + C | Z)}\right].$$

Importantly, the righthand term includes $P(T|Z)$ in the denominator. As T becomes very predictable from Z, these probabilities go to either 1 or 0. For the terms where the denominator goes to 0, they diverge, meaning that no regular estimator can achieve a finite variance.

D Details on Debiased Machine Learning

D.1 Fitting the Outcome Models

As we discuss in section 5.1, we select 10 proxies that have the strongest relationship with the outcome variable based on a univariate cross-validated kernel ridge regression model. We then train our models on all combinations of these ten chosen candidate features. For each sublist, we also include either age and cohort trends or age and cohort dummies. We therefore train our models on $2 \times (2^{10} - 1) = 2046$ different feature matrices for each country.

For each feature matrix, we remove each column's mean and rescale by the standard deviation. For some highly-skewed features we remove the median and rescale by the interquartile range instead (using sklearn's RobustScaler). We train various models over a range of hyperparameters. We use log-spaced grids for hyperparameters. Denote an evenly spaced grid of n elements between endpoints x_0 and x_1 as: $g(x_0, x_1, n)$. Let log_grid(x_0, x_1, n) be the grid containing 10 to the power of i for $i \in g(\log x_0, \log x_1, n)$. We train the following models:

- Ridge regression with hyperparameters $\lambda \in$ log_grid(1e-7, 20, 100)

- Lasso with hyperparameters $\lambda \in \text{log_grid}(\text{1e-7}, 20, 100)$
- Kernel ridge regression with $\lambda \in \text{log_grid}(\text{1e-7}, \text{2e-1}, 20)$, $\gamma \in \text{log_grid}(\text{1e-6}, \text{1e-1}, 10)$, with both RBF and Laplacian kernels.

We use an evenly spaced grid in logs because the variance can grow very rapidly as the regularization parameters near 0, so we want the grid points to be closer together near 0. This is standard practice in machine learning.

Finally, note that we choose relatively coarse grids with only 100 to 400 different combinations of hyperparameters per method. This saves computation in our initial pass that performs this procedure on different combinations of feature matrices. However, after we have found the best combination of feature matrix and hyperparameter for ridge/lasso/KRR, we then go back and refine the hyperparameters on only these feature matrices with finer grids containing 10x more points. This coarse-to-fine strategy is also standard practice in machine learning.

D.2 The Choice of Machine Learning Estimators

As we describe in Section 5.1, we fit each machine learning algorithm (ridge, kernel ridge, etc) on every combination of features. With unlimited compute, we would run any conceivable machine learning algorithm, including tree-ensembles and neural networks, on each of the feature matrices. However, in practice, we find that given our very small number of sample sizes (in the 100s) more complicated methods like tree-ensembles and neural networks have two problems: (1) they perform poorly across the board compared to simpler models, and (2) they take much longer to run. Therefore, we exclude these more complicated models from our full run.

To demonstrate these two claims, in this section we present results with tree-ensembles and neural networks on the GRID data for the United States. We begin by selecting the features that yielded the best results under our main specification. These are: Age, Cohort, Square-root of Age, Log Oil Prices, Lagged Log Oil Prices, and Lagged Log Industrial Production Squared. The best performing model from our main specification is Kernel Ridge Regression, using the Laplacian kernel with bandwidth 0.0328 and regularization penalty 1.86×10^{-5}. This model achieves a cross-validated root mean squared error of 0.013. We compare this result to several tree and neural network specifications in Table D.1.

The much simpler kernel ridge regression method achieves *nearly half* the RMSE of the more complicated models. Note that the typical neural network used for regression in small datasets, the Multilayer Perceptron (MLP) performs particularly poorly. RealMLP (Holzmüller et al., 2024)

Table D.1: Comparison of Tree and Neural Network Models to Kernel Ridge Regression

Machine Learning Method	Cross-validated RMSE
Kernel Ridge Regression	**0.013**
Random Forest	0.022
Gradient-Boosted Trees	0.021
Multilayer Perceptron	0.111
RealMLP	0.021

Note: Inputs to the models are: Age, Cohort, Square-root of Age, Log Oil Prices, Lagged Log Oil Prices, and Lagged Log Industrial Production Squared. For cross-validation, we use year-blocked splits of 8 contiguous years.

is a recent improved architecture for small datasets that incorporates an extensive series of tricks and transformations — even this state-of-the-art neural architecture performs poorly in this setting compared to kernel ridge.

Finally, kernel ridge regression tends to suffer in high dimensions. This is one reason why we have to iterate through all possible subsets of the original features to find the best model. Tree ensembles and neural networks on the other hand typically perform well with high-dimensional inputs. Therefore, we also test these models using all available features including all transformations of age, cohort, and all proxy variables. For the GRID US data, this is 53 features. The results are presented in Table D.2.

Table D.2: Tree and Neural Network Models with High-Dimensional Input

Machine Learning Method	Cross-validated RMSE
Random Forest	0.023
Gradient-Boosted Trees	0.024
Multilayer Perceptron	0.722
RealMLP	0.025

Note: Models are trained using all 53 input features (excluding only those proxies with an $R^2 > 0.75$ in the regression of calendar year on that proxy). For cross-validation, we use year-blocked splits of 8 contiguous years.

Reflecting the difficulties of the very small sample size, the results are always worse than what was achieved in the more carefully selected subset of features in Table D.1.

D.3 Fitting the Weighting Models

In this section, we provide details on how we estimate the weights $\hat{\alpha}$. In general, these weights are found by minimizing the loss from Chernozhukov et al. (2022):

$$\min_{\alpha \in \mathcal{F}} \left\{ E[\alpha(A, C, Z)^2 - 2 \cdot \text{slope}(\alpha)] \right\}. \tag{14}$$

In the most general case, $\mathcal{F}$ can be an arbitrary function class. Different choices of $\mathcal{F}$ will result in different estimates with different properties. For the applications used in Chernozhukov et al. (2022), $\mathcal{F}$ is the set of linear functions whose coefficients are within an ℓ_1-norm ball — just as in the Lasso. See Chernozhukov et al. (2022) for details on the "Lasso" type weights.

In this work, we consider two function classes that can be solved analytically: linear functions with coefficients in an ℓ_2-norm ball (ridge weights), and functions within an RKHS ball (kernel ridge weights). Access to a closed-form solution offers substantial computational advantages. These ridge-type estimators of the Riesz representer have been studied theoretically in Singh (2021).

In the subsections below, we walk through the derivation of the closed-form solutions for ridge weights and kernel ridge weights.

But other than the loss function, we estimate the weights with exactly the same machine learning setup as for our basic prediction problem. This includes both the choice of features, and using blocked cross-validation for model selection. We train models for the weights with the following hyperparameters:

1. Ridge weights with $\delta \in \text{log_grid}(\text{1e-7}, \text{5e-1}, 25)$.
2. Kernel Ridge weights with $\delta \in \text{log_grid}(\text{1e-7}, \text{5e-1}, 20)$, $\gamma \in \text{log_grid}(\text{1e-6}, \text{1e-1}, 10)$, with the RBF kernel.

As in Appendix D.1, we refine the hyperparameters on a finer grid for the best performing feature matrix / model combinations.

D.3.1 Linear Weights

First, we consider linear function classes with ℓ_2-norm regularization, aka Ridge for weighting. Write $X(A, C, Z) \in \mathbb{R}^{n \times d}$ for a design matrix with a row for each observation, and columns for each feature. For example, we might have $d = 3$ with simple linear features A, C, and Z. Or we might have $d = 5$ with features A, A^2, C, C^2, Z. Note that we have written X as a function of A, C, Z. This will be important later when computing the optimal weights. When this dependence on A, C, Z is not important for the context, we simply write X for brevity. Without loss of generality, we assume that X has mean zero, otherwise we can just recenter first.

The loss in the linear case is:

$$\min_{\beta \in \mathbb{R}^d} \{ \frac{1}{n} \beta^\top X^\top X \beta - 2\text{slope}(X\beta) + \delta \beta^\top \beta \}, \tag{15}$$

where δ is a hyperparameter. To get the optimal weights, we exploit the fact that because slope is a linear functional, there must be a way to write the slope as a linear function of β. We derive this form next.

Recall that our estimand is the slope of the age profile. Then by straightforward OLS math, the slope of the age profile is a linear combination of the elements of the age profile. In particular, recall that the value of the age profile at age a is $\sigma_a^2 = \mathbb{E}[m(a, C, Z)]$. Let $\sigma_{\vec{a}}^2$ be the full age profile as a vector whose elements are σ_a^2. Furthermore, define $\vec{a} \coloneqq \{a_{\text{min}}, ..., a_{\text{max}}\}$ and let $\bar{a}$ denote the mean of $\vec{a}$. Then, we have:

$$\text{slope} = \frac{(\vec{a} - \bar{a})^\top \sigma_{\vec{a}}^2}{(\vec{a} - \bar{a})^\top (\vec{a} - \bar{a})}.$$

Denote:

$$s \coloneqq \frac{(\vec{a} - \bar{a})}{(\vec{a} - \bar{a})^\top (\vec{a} - \bar{a})}.$$

Notice that for m of the form $X\beta$ we have $\sigma_a^2 = \mathbb{E}[\beta^\top X(a, C, Z)]$. Putting this all together:

$$\begin{aligned} \text{slope}(X\beta) &= s^\top \mathbb{E}[\beta^\top X(\vec{a}, C, Z)] \\ &= \beta^\top \mathbb{E}[s^\top X(\vec{a}, C, Z)], \end{aligned}$$

where $\mathbb{E}[s^\top X(\vec{a}, C, Z)] =: \bar{X}_{\text{cf}} \in \mathbb{R}^d$ are the average "counterfactual features," to borrow the language from Chernozhukov et al. (2020). Note that when we are solving for the weights in practice, the expectation in the definition of $\bar{X}_{\text{cf}}$ will be the sample average.

Now we are ready to get a closed-form solution to Equation (15). Using the first order conditions, we have the coefficients for the optimal weights:

$$\beta^* = (X^\top X/n + \delta I)^{-1} \bar{X}_{\text{cf}}.$$

These closed-form coefficients can be computed quickly in just a few lines of code. The first term, $(X^\top X/n + \delta I)^{-1}$ is just the standard inverse regularized covariance matrix from ridge regression. The only new term we must compute is $\bar{X}_{\text{cf}}$. For each observation, we have to compute the counterfactual features $X(a, C_i, Z_i)$ and take their inner product with s. Then we just take the sample average of the result.

The final estimated weighting function is:

$$\hat{\alpha}(x) = x^\top \beta^*.$$

D.3.2 Kernel Weights

We adopt the same setup with a feature matrix X as for linear weights, but now we solve the weighting optimization problem over a Reproducing Kernel Hilbert Space (RKHS) — see Scholkopf and Smola (2018) for a review. We broadly adopt the setup from Hirshberg et al. (2019); Singh (2021); Bruns-Smith et al. (2023). Let $\mathcal{H}$ be a possibly-infinite-dimensional RKHS with kernel $\mathcal{K}$. Let $\|\cdot\|_{\mathcal{H}}$ denote the norm of $\mathcal{H}$. Let K be the matrix with entries $\mathcal{K}(x_i, x_j)$, where x_i, x_j are the ith and jth entries of X. Then the weighting problem is:

$$\min_{a \in \mathbb{R}^n} \{\frac{1}{n}\alpha^\top K^\top K\alpha - 2\text{slope}(K\alpha) + \delta\alpha^\top K\alpha\}$$

where

$$\text{slope}(K\alpha) = \alpha^\top \bar{K}_{\text{cf}},$$

and the counterfactual kernel matrix $\bar{K}_{\text{cf}}$ has a closed-form. Define K_a to be the matrix with entries $\mathcal{K}(x_i, x'_j)$ where x_i is the ith row of $X(A, C, Z)$ and x'_j is the (counterfactual) jth row of $X(a, C, Z)$. Define $\bar{K}_a \in \mathbb{R}^n$ to be the average for each row of K_a. Define $\bar{K}_{\vec{a}} \in \mathbb{R}^{n \times n_{\text{age}}}$ to be the matrix whose columns are the $\bar{K}_a$ for each a. Then:

$$\bar{K}_{\text{cf}} \coloneqq \bar{K}_{\vec{a}} s,$$

for s defined as in Appendix D.3.1. This follows directly from algebra using the definition of the slope applied to the kernel ridge model with dual coefficients α.

There is a closed-form solution to the optimization problem:

$$\alpha^* = (K^T K/n + \delta K)^{-1} \bar{K}_{\text{cf}}.$$

For a single point x, define K_x as the vector with entry $\mathcal{K}(x_i, x)$ for each row x_i in X. Then the resulting estimated weighting function is:

$$\hat{\alpha}(x) = K_x^\top \alpha^*.$$

D.4 Standard Errors for the Slope

We now describe how to compute asymptotic normal standard errors for the debiased machine learning estimate. We do this in the usual way taken directly from Chernozhukov et al. (2022), but

give details here for concreteness.

Let slope$(\hat{m}, c, z)$ denote the slope of model $\hat{m}$ evaluated at a fixed c and z. In other words, we take the age profile at a fixed c and z, $\hat{m}(35, c, z), \hat{m}(36, c, z), ..., \hat{m}(50, c, z)$, and then compute the slope of the best fit line through this profile. Then by linearity, we have slope$(\hat{m}) = E[\text{slope}(\hat{m}, c, z)]$.

Then our debiased point estimate using predictive model $\hat{m}$ and Riesz representer model $\hat{\alpha}$ can be written:

$$\hat{\theta}_{\text{dr}} \coloneqq \frac{1}{n}\sum_{i=1}^{n}\Big\{\text{slope}(\hat{m}, c_i, z_i) + \hat{\alpha}(a_i, c_i, z_i)(\sigma_i^2 - \hat{m}(a_i, c_i, z_i))\Big\} \tag{16}$$

This is a sample average, and so the asymptotic variance is estimated in the usual way:

$$\hat{V}_{\text{dr}} \coloneqq \frac{1}{n}\sum_{i=1}^{n}\Big\{\big(\text{slope}(\hat{m}, c_i, z_i) + \hat{\alpha}(a_i, c_i, z_i)(\sigma_i^2 - \hat{m}(a_i, c_i, z_i)) - \hat{\theta}_{\text{dr}}\big)^2\Big\}$$

and our asymptotic normal standard error is $\sqrt{\hat{V}_{\text{dr}}/n}$.

This contrasts with the biased machine learning plug-in estimator:

$$\hat{\theta}_{\text{ml}} \coloneqq \frac{1}{n}\sum_{i=1}^{n}\Big\{\text{slope}(\hat{m}, c_i, z_i)\Big\}$$

and the naive standard errors that would result from erroneously assuming that $\hat{\theta}_{\text{ml}}$ is unbiased:

$$\hat{V}_{\text{ml}} \coloneqq \frac{1}{n}\sum_{i=1}^{n}\Big\{\big(\text{slope}(\hat{m}, c_i, z_i) - \hat{\theta}_{\text{ml}}\big)^2\Big\}$$

The standard errors using $\hat{V}_{\text{ml}}$ will not generally yield a confidence interval with the desired coverage, as we illustrate in simulation in Appendix G.

Note that the debiased variance is inflated by the inclusion of the weights and residuals. Ignoring cross-terms, we can use the following rough bound for intuition:

$$\begin{aligned}\hat{V}_{\text{dr}} &\approx \hat{V}_{\text{ml}} + \frac{1}{n}\sum_{i=1}^{n}\hat{\alpha}(a_i, c_i, z_i)^2(\sigma_i^2 - \hat{m}(a_i, c_i, z_i))^2 \\ &\leq \hat{V}_{\text{ml}} + \underbrace{\left(\sum_{i=1}^{n}\hat{\alpha}(a_i, c_i, z_i)^2\right)}_{\text{size of }\hat{\alpha}}\underbrace{\left(\frac{1}{n}\sum_{i=1}^{n}(\sigma_i^2 - \hat{m}(a_i, c_i, z_i))^2\right)}_{\text{MSE of }\hat{m}}.\end{aligned}$$

In other words, the difference between the debiased confidence interval and the naive confidence interval depends roughly on (1) the sum of the squared weights, and (2) the mean squared error of the predictor $\hat{m}$. If the predictor $\hat{m}$ were perfect, we wouldn't need to perform any bias correction. If the predictor is imperfect, then intuitively the weights $\hat{\alpha}$ control how the uncertainty from the predictions passes through to our uncertainty about $\hat{\theta}_{\text{dr}}$.

D.5 Cross-Fitting

Notice that the added bias correction term in Equation (16) takes the form:

$$\frac{1}{n}\sum_{i=1}^{n}\Big\{\hat{\alpha}(a_i, c_i, z_i)(\sigma_i^2 - \hat{m}(a_i, c_i, z_i))\Big\}. \tag{17}$$

When the prediction residuals $\sigma_i^2 - \hat{m}(a_i, c_i, z_i)$ are small, then the impact of bias correction is also small. This can be an issue if we use the data to fit the model and then subsequently use the same data to calculate the residuals. In the worst-case, we could severely overfit, driving the residuals to zero. In practice, we select $\hat{m}$ using blocked cross-validation, so we never drive the residuals to zero. However, there could be a concern that the residuals would still be too small, since we use the same data to select $\hat{m}$ and to compute the residuals. This has been an area of active debate in the debiased machine learning literature. Theoretically, when the machine learning algorithm used to select $\hat{m}$ satisfies a "Donsker condition" (van der Vaart and Wellner, 2023), then reusing the samples is not a problem. However, the Donsker condition essentially requires that the algorithm cannot be too flexible, which may be violated by many modern machine learning tools. Therefore many debiased machine learning papers (e.g. see Chernozhukov et al., 2018) suggest using "cross-fitting", where the models are trained in folds and then applied pseudo-out-of-sample as with cross-validation. A recent simulation study in Bach et al. (2024) finds that using the whole sample or doing cross-fitting perform roughly the same, but to guarantee that our confidence intervals are valid, we choose to use cross-fitting for our final estimates. Unfortunately, this does make notation a bit messier, and so we suppress the details from the main text and present them below.

We define folds according to our blocked cross-validation scheme. Denote each of these test folds k. Let K be the total number of folds. Let I_k denote the indices of the observations within fold k and let $I_{\neg k}$ denote the indices of observations outside of fold k. Let $\hat{m}_k$ and $\hat{\alpha}_k$ be the predictors and weights fit on the data in $I_{\neg k}$. Let $X_i \coloneqq \{A_i, C_i, Z_i\}$ denote the data for observation i.

Then the cross-fit point estimate is:

$$\hat{\theta}_{\text{dr-cf}} \coloneqq \frac{1}{K}\sum_{k=1}^{K}\frac{1}{|I_k|}\sum_{i\in I_k}\Big\{\text{slope}(\hat{m}_k, c_i, z_i) + \hat{\alpha}_k(a_i, c_i, z_i)(\sigma_i^2 - \hat{m}_k(a_i, c_i, z_i))\Big\}, \tag{18}$$

and the cross-fit variance is:

$$\hat{V}_{\text{dr-cf}} \coloneqq \frac{1}{K}\sum_{k=1}^{K}\frac{1}{|I_k|}\sum_{i\in I_k}\Big\{\big(\text{slope}(\hat{m}_k, c_i, z_i) + \hat{\alpha}_k(a_i, c_i, z_i)(\sigma_i^2 - \hat{m}_k(a_i, c_i, z_i)) - \hat{\theta}_{\text{dr-cf}}\big)^2\Big\}. \tag{19}$$

Finally, we compute the standard errors:

$$\widehat{\text{std err}}_{\text{cf}} = \sqrt{\hat{V}_{\text{dr-cf}}/n}\,. \tag{20}$$

Repeated cross-fitting and median aggregation. The estimates above depend on the particular random partition of the years into folds. To reduce this dependence, we repeat the cross-fitting procedure S times, each time with a newly drawn random offset for the circular blocks. Repetition s yields a point estimate $\hat{\theta}_s$ and variance $\hat{V}_s$, computed as in (18) and (19). Following Chernozhukov et al. (2018), we aggregate across repetitions with the median:

$$\hat{\theta} \coloneqq \text{median}_{s=1,\dots,S}\ \hat{\theta}_s, \qquad \hat{V} \coloneqq \text{median}_{s=1,\dots,S}\Big\{\hat{V}_s + \big(\hat{\theta}_s - \hat{\theta}\big)^2\Big\}. \tag{21}$$

The term $(\hat{\theta}_s - \hat{\theta})^2$ accounts for the variation due to the random choice of split. The final standard error is $\sqrt{\hat{V}/n}$.

D.6 Clustered Standard Errors

Our blocked cross-validation scheme splits the data by whole years. This guarantees that observations from the same year (which are strongly dependent) do not simultaneously appear in both a training set and test set. For the same reason, we also extend (19) to handling clustering at the year level.

Let $I_{k,t}$ be the index set of observations in year t and fold k. We compute the squared cluster sums:

$$g_{k,t}^2 = \left(\sum_{i\in I_{k,t}}\big(\text{slope}(\hat{m}_k, c_i, z_i) + \hat{\alpha}_k(a_i, c_i, z_i)(\sigma_i^2 - \hat{m}_k(a_i, c_i, z_i)) - \hat{\theta}_{\text{dr-cf}}\big)\right)^2$$

Let g_t^2 denote the average of $g_{k,t}^2$, averaging over all folds.

Then our estimate of the asymptotic variance (including finite sample correction for our number of clusters, n_{year}) is:

$$\hat{V}_{\text{dr-cluster}} := \left(\frac{n_{\text{year}}}{n_{\text{year}} - 1}\right) \frac{1}{n} \sum_t g_t^2.$$

Over repeated random splits of the data into folds, we perform median aggregation as described above.

E Debiased Estimate of the Age Profile

E.1 Estimating the Full Age Profile

In the main text and in Appendix D, we have discussed estimation of the slope of the age profile. Beyond estimating the slope, we can also debias each point σ_a^2 on the age profile individually to obtain a full debiased age profile. For fixed values of the hyperparameter and proxies used for the weights, the two are *equivalent*. That is, if we were to debias each point individually, and then compute the slope of the resulting age profile, this is numerically equivalent to the debiased estimate of the slope.

We use this fact to plot a corresponding debiased age profile for each of our slope point estimates — these are what we plot in Figure 5 and Figure 7, and what we use to get the "Debiased ML" columns of Table 2 and Table 3. We take the cross-validated kernel weighting hyperparameters and proxy variables that are selected for the slope, and use them to debias each point on the age profile individually like so:

Consider a fixed point on the age profile. For concreteness, we'll use σ_{35}^2, the value of the age profile at age 35. The Riesz loss problem for debiasing σ_{35}^2 is:

$$\alpha_{35}^* = \underset{\alpha}{\text{argmin}} \big\{ E[\alpha(a, c, z)^2 - 2 \cdot \alpha(35, c, z)] \big\}. \tag{22}$$

We can then run exactly the same debiased machine learning procedure that we outlined for the slope, but for σ_{35}^2. The prediction problem for σ_{at}^2 is the same, so we will use the same predictive model, $\hat{m}$. If we have a Riesz representer model $\hat{\alpha}_{35}$, the resulting debiased estimate of the age profile at age 35 is:

$$\hat{\sigma}_{35}^2 := \frac{1}{n} \sum_{i=1}^{n} \Big\{ \hat{m}(35, c_i, z_i) + \hat{\alpha}_{35}(a_i, c_i, z_i)(\sigma_i^2 - \hat{m}(a_i, c_i, z_i)) \Big\} \tag{23}$$

The slope of the resulting profiles are numerically identical to the reported slope estimates in Table 1 and Table 4.

E.2 Proof of the Equivalence

For completeness, we provide a derivation of the equivalence, described in the previous section, between the debiased estimate of the slope and the slope of a debiased estimate of the age profile.

We will write $\vec{a}$ to indicate we mean the vector of values we get from setting age equal to $35, 36, ..., 50$. So $\hat{\sigma}^2_{\vec{a}}$ is the whole debiased estimated age profile following the form of $\hat{\sigma}^2_{35}$ in Equation (23). Using this notation, the slope through this estimated age profile would be:

$$\begin{aligned}\text{slope}(\hat{\sigma}^2_{\vec{a}}) &= \text{slope}\left(\frac{1}{n}\sum_{i=1}^{n}\left\{\hat{m}(\vec{a}, c_i, z_i) + \hat{\alpha}_{\vec{a}}(a_i, c_i, z_i)(\sigma_i^2 - \hat{m}(a_i, c_i, z_i))\right\}\right) \\ &= \frac{1}{n}\sum_{i=1}^{n}\left\{\text{slope}(\hat{m}, c_i, z_i) + \text{slope}(\hat{\alpha}_{\vec{a}}(a_i, c_i, z_i))(\sigma_i^2 - \hat{m}(a_i, c_i, z_i))\right\}. \end{aligned} \tag{24}$$

We are relying on the fact that the slope is calculated as a linear combination, and therefore can move inside the sum. Compare Equation (24) to the debiased point estimate for the slope from Equation (16). The only difference is that while Equation (16) has a single weighting model for the slope $\hat{\alpha}$, in Equation (24), we have the combination of individual weighting models, $\text{slope}(\hat{\alpha}_{\vec{a}}(a_i, c_i, z_i))$.

These turn out to be equivalent in finite sample for the linear and kernel weighting models we consider in Appendix D.3. We provide the derivation for the linear case below (the argument for the kernel setting is virtually identical).

We first derive the solution to the linear weighting loss for a fixed point on the age profile (again, we use age 35 as an example). Let X_{35} denote the covariates where all ages are set to equal 35, and write $\bar{X}_{35} \in \mathbb{R}^d$ for the sample average of the columns of X_{35}. Then the weighting problem is:

$$\min_{\beta \in \mathbb{R}^d}\left\{\frac{1}{n}\beta^\top X^\top X\beta - 2\bar{X}_{35}^\top\beta + \delta\beta^\top\beta\right\}. \tag{25}$$

Taking the derivative with respect to β, the solution is:

$$\beta^*_{35} = (X^\top X/n + \delta I)^{-1}\bar{X}_{35},$$

with corresponding weights $\hat{\alpha}_{35} = X\beta^*_{35}$. Then following the same argument from Appendix D.3.1,

for

$$s \coloneqq \frac{(\vec{a} - \bar{a})}{(\vec{a} - \bar{a})^\top (\vec{a} - \bar{a})},$$

the slope through the weights are:

$$\begin{aligned}
\text{slope}(X\beta^*_{\vec{a}}) &= (X\beta^*_{\vec{a}})s \\
&= X(X^\top X/n + \delta I)^{-1}(\bar{X}_{\vec{a}} s) \\
&= X(X^\top X/n + \delta I)^{-1}\bar{X}_{\text{cf}},
\end{aligned}$$

which is exactly the solution to the weights for the slope derived in Appendix D.3.1.

Note that repeated cross-fitting may introduce some small amount of noise to this numerical equivalence. If we use the mean of the point estimates to aggregate across repeated splits, then the numerical equivalence holds exactly. However, if we use the median (which is more standard following Chernozhukov et al. (2018)) then there will be a small finite sample difference due to nonlinearity of the median.

F GMM Details

In the cross-sectional variance of income application, we estimate the persistence parameter ρ using GMM. We use our estimated age profiles as inputs to the GMM estimation for the parameters of the following income process:

$$y_{i,a} = \alpha_i + z_{i,a} + \epsilon_{i,a} \tag{26}$$

$$z_{i,a} = \rho z_{i,a-1} + \eta_{i,a}. \tag{27}$$

Recall that we assume for simplicity that the processes are centered around zero,i.e., $E[y_{i,a}] = E[z_{i,0}] = E[\alpha_i] = E[\epsilon_{i,a}] = E[\eta_{i,a}] = 0$. We denote the variances of the random variables as $\sigma^2_\alpha, \sigma^2_\epsilon$, and σ^2_η. We write the initial permanent variance as $\sigma^2_{z0} \coloneqq \text{Var}[z_{i,0}]$.

The relationship between the age profile and the model parameters is as follows:

$$\begin{aligned}
\sigma^2_a &= E[y^2_{i,a}] \\
&= \sigma^2_\alpha + E[z^2_{i,a}] + \sigma^2_\epsilon \\
&= \sigma^2_\alpha + (\rho^2)^a \sigma^2_{z0} + \sum_{k=1}^{a} (\rho^2)^{a-k} \sigma^2_\eta + \sigma^2_\epsilon.
\end{aligned} \tag{28}$$

Note that we begin indexing at $a = 0$, which corresponds to age 35 in our baseline working age specification.

The full set of parameters $\rho, \sigma^2_\alpha, \sigma^2_\epsilon, \sigma^2_\eta$, and σ^2_{z0} are not identified from the age profile alone. For this we would need to add covariance moments. However, ρ is pinned down by the curvature of the age profile.

We estimate ρ jointly with $\sigma^2_\eta, \sigma^2_{z0}$, and $\sigma^2_\alpha + \sigma^2_\epsilon$ using the scipy.optimize library in python. We minimize the (equally weighted) sum of squares of the errors to the estimated data moments, i.e. the age profile, $\hat{\sigma}^2_a$. For computational convenience, we impose the bounds: $\sigma^2_\eta, \sigma^2_{z0}, \sigma^2_\epsilon + \sigma^2_\alpha \in [0, 2]$ and $\rho \in [0.01, 2]$. These constraints never bind. We run the solution algorithm 200 times using uniform random initializations of the parameters within the bounds. The estimates for ρ are highly stable across these runs.

In the case where we estimate the age profile $\hat{\sigma}^2_a$ using our debiased machine learning method (with predictive model $\hat{m}$ and weighting model $\hat{w}$), we can also get a valid confidence interval for the GMM estimate of ρ. We do this using the standard formula:

$$\hat{V}_{\text{gmm}} = (D^\top D)^{-1} D^\top \Sigma D (D^\top D)^{-1},$$

where $D \in \mathbb{R}^{n_{\text{age}} \times 4}$ is the Jacobian of the GMM errors with respect to the parameters evaluated at the GMM solution, and $\Sigma \in \mathbb{R}^{n_{\text{age}} \times n_{\text{age}}}$ is the covariance of the errors. See for example, Cameron and Trivedi (2005) for further details. We numerically approximate D with a centered second-order finite difference calculation.

When estimating Σ, we make sure to propagate the uncertainty involved in estimating the age profile using the debiased machine learning estimating equation. Each individual element of the age profile is estimated using equation (23). As described in Appendix E, we estimate the individual weights using the same model we selected via blocked cross-validation for the slope weights. Let $\sigma^2_{\text{gmm}} \in \mathbb{R}^{n_{\text{age}}}$ denote the age profile implied by plugging in the GMM-estimated parameters into equation (28). Then the GMM error vector calculated at a single observation i is:

$$\text{err}_i := \hat{m}(\vec{a}, c_i, z_i) + \hat{\alpha}_{\vec{a}}(a_i, c_i, z_i)(\sigma^2_i - \hat{m}(a_i, c_i, z_i)) - \sigma^2_{\text{gmm}},$$

with $\text{err}_i \in \mathbb{R}^{n_{\text{age}}}$, and the error covariance matrix is:

$$\frac{1}{n} \sum_{i=1}^{n} \text{err}_i \text{err}_i^\top.$$

Let the diagonal entry of $\hat{V}_{\text{gmm}}$ that corresponds to the persistence parameter be denoted $\hat{V}_\rho$. Then the standard error for our estimated persistence is $\sqrt{\hat{V}_\rho/n}$. In practice, we do the variance and standard error calculations using the cross-fitting procedure in Appendix D.5 to avoid underestimating the uncertainty.

Finally, note that we can obtain clustered standard errors by correcting for clustering in the construction of Σ. Let $I_{\ell,t}$ be the index set of observations in year t and fold ℓ. We compute the cluster sums:

$$g_t = \sum_{\ell} \sum_{i \in I_{\ell,t}} \text{err}_i.$$

Let G be the matrix in $\mathbb{R}^{n_{\text{year}} \times n_{\text{age}}}$ whose rows are the g_t for each year. Then the clustered error covariance is:

$$\Sigma = \left(\frac{n_{\text{year}}}{n_{\text{year}} - 1} \right) \frac{1}{n} G^\top G.$$

We then compute $\hat{V}_{\text{gmm}}$ as above.

G Simulation Study

We perform a simulation study to validate our debiased proxy machine learning methodology. In this study we illustrate: (1) the role of the proxy variables in identification, and (2) the importance of the debiasing step for obtaining a valid confidence interval. A key concern is that bias arising from regularization may be the reason why our machine learning estimates yield a small value for the slope of the age profiles we estimate. To investigate this possibility, we construct a simulation that serves as a placebo test.

For concreteness, the simulation we perform is meant to assess our analysis of the cross-sectional variance of income. On the real US data, the slope of the age profile estimated by the age-cohort model is larger than the slope of the age profile estimated by the age-time model. To illustrate that this is not a mechanical consequence of our procedure, we design our simulation such that the steeper age profile is actually the truth (the age-cohort slope is close to the true slope), and we want our debiased proxy machine learning approach to reliably estimate this high slope. This way we can demonstrate that the small estimate for the age slope in the US is not due to regularization bias.

G.1 DGP Calibration

We use a data-generating process (DGP) that is calibrated to the US GRID data. We use the same age range and same number of years as in the real data. We fix the slope of the true age profile over ages 35–50 to 0.9%.[15] This value is about three times larger than our debiased ML estimate for the US GRID data. To construct the DGP, we first use a kernel ridge regression to regress σ^2_{at} in the US GRID data on age, the square root of age, cohort, log oil price, lagged log oil price, and lagged log industrial production squared. We refer to the resulting predictive model as $\hat{f}(a, c, z_t)$; its implied age slope is $\theta_0 = 0.36\%$. We then raise the slope by adding $\Delta \cdot a$, where $\Delta := 0.9\% - \theta_0$. To keep the simulated data close to the real data, we also subtract $\hat{g}(c, z_t)$, a kernel ridge regression of $\Delta \cdot a$ on cohort and the proxies from the real data, which removes as much of the added age trend as can be explained without age. This gives the mean function

$$f(a, c, z_t) := \hat{f}(a, c, z_t) + \Delta \cdot a - \hat{g}(c, z_t),$$

which retains the interactions between age, cohort, and the proxies present in the real data. Because $\hat{g}$ does not depend on age, the true age slope is $\theta_0 + \Delta = 0.9\%$.

For each simulation draw, we redraw the proxies z_t and the outcomes σ^2_{at}, resulting in a new dataset with $n = 682$ observations. We redraw the proxies by adding iid normal noise to each original annual proxy series. The standard deviation of the noise is set to be $1/3$rd the standard deviation of the corresponding original series. Lagged and squared proxies are recomputed from the perturbed series. We redraw the outcomes as:

$$\sigma^2_{at} = f(a, c, z_t) + u_t + \nu_{at},$$

where u_t is a mean zero iid normal year shock with standard deviation 0.002, and ν_{at} is mean zero iid normal noise with standard deviation 0.0018. We calibrate the latter from the second differences of σ^2_{at} across adjacent ages within a year in the real data, which removes year-level variation. The year shocks induce dependence within years, which our clustered standard errors are designed to handle. To mimic proxy selection, we also add two decoy proxies that do not enter the DGP: log GDP growth and squared log consumption growth, the next-best proxies in the univariate ranking for the US. These are redrawn in the same way.

[15] Throughout this section, as in the main text, we report the slope values multiplied by 100 and designate this with %. The original values are slopes in log units, and so this is a slight abuse of notation.

Table G.1: Simulation Study Results

Estimator	Avg. Estimate	Bias	RMSE	Coverage	
Debiased Machine Learning	0.898	−0.002	0.110	94.7%	[93.1%, 96.1%]
Biased Machine Learning	0.670	−0.231	0.234	0.0%	[0.0%, 0.37%]

Note: Results from 1000 simulation draws. We present the slopes multiplied by 100 to ease readability. The true slope of the age profile over ages 35–50 is 0.900. Bias, RMSE, and coverage are computed relative to each draw's true slope. Coverage is for nominal 95% confidence intervals with standard errors clustered by year. The biased ML interval is the naive interval without debiasing. In brackets, we give the exact Clopper-Pearson 95% interval on coverage with respect to the sampling variability of the 1000 trials.

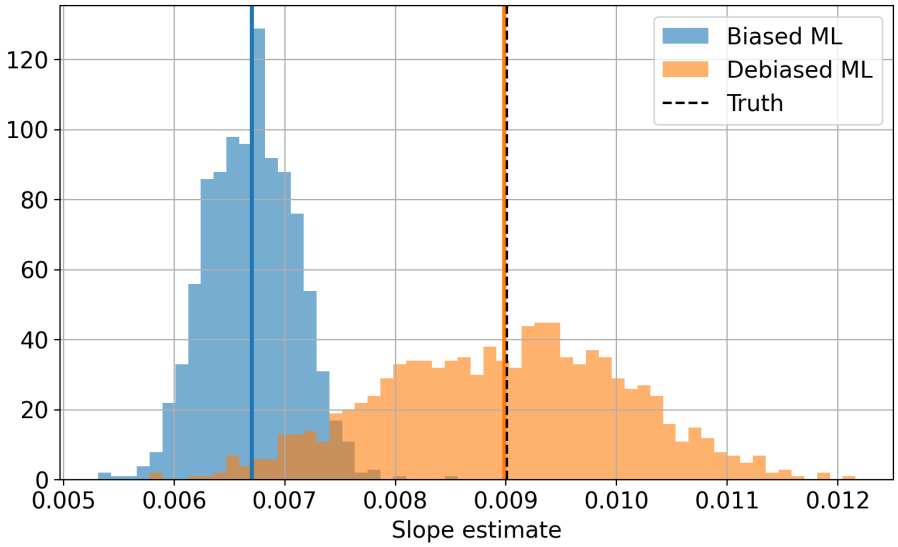


Figure G.1: Simulation Point Estimates vs the Truth

Note: The figure presents histograms of the biased ML and debiased ML point estimates of the slope of the age profile over 1000 draws. Each histogram has a corresponding vertical line for the mean point estimate. The dotted vertical line is the ground truth slope.

G.2 Simulation Results

For each draw from the DGP described above, we run our proxy machine learning methodology with kernel ridge regression for the predictive model and kernel balancing weights for the weighting model. In each draw, we select the predictive model, both the set of proxies and the ridge penalty, with blocked cross-validation, as we do with the real data. We choose among the true proxies, the true proxies plus one or both decoys, and each pair of true proxies. We compute the debiased point estimate and confidence interval as described in the main text, using $K = 3$ contiguous year blocks for cross-fitting, eight repeats of block splitting aggregated by the median, and standard errors clustered by year. For illustrative purposes, we also compute a naive confidence interval for the machine learning estimate without debiasing, applying the same clustered standard error formula to the plug-in estimate.

We repeat this process for 1000 draws. Figure G.1 plots the distribution of point estimates. We summarize the results in Table G.1. Notice that the biased ML estimate catastrophically undercovers: in 0 out of 1000 draws does the naive confidence interval cover the true slope. The average biased ML estimate is 0.67%, well below the true slope. In other words, without debiasing, ML regularization results in a substantially biased estimate of the slope. By contrast, our debiased procedure achieves coverage of 94.7%, within sampling variation of the nominal 95%. We achieve this improved coverage largely because debiasing essentially eliminates the bias: the bias of the debiased estimate is -0.002%, statistically indistinguishable from zero, compared with -0.231% for the biased ML estimate. Notice from Figure G.1 that the debiased standard errors are larger, which contributes to improved coverage. Despite the increase in variance (which more accurately represents the underlying uncertainty in the point estimate), debiasing also cuts the root mean squared error in half.

H Additional Results for Cross-Sectional Variance of Income

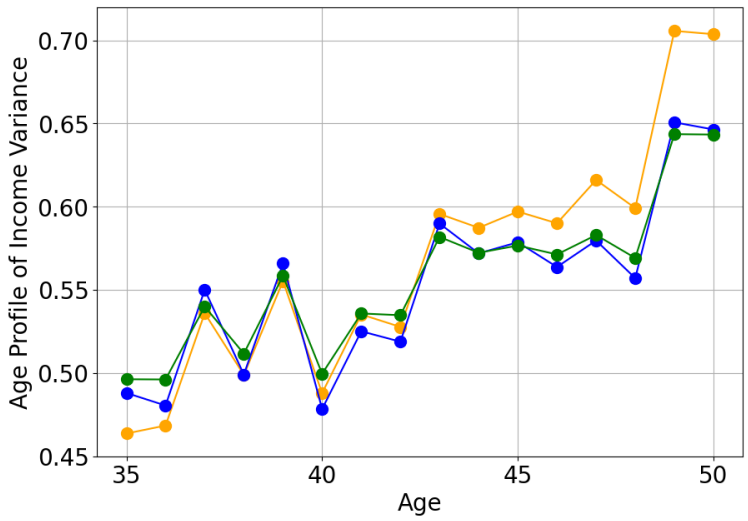


Figure H.1: Age Profile of Cross-Sectional Income Variance in the US

Note: The figure plots the estimated age profile of the cross-sectional variance of income for three models using PSID data.

Table H.1: International Persistence, ρ.

Country	Years	Age-Cohort	Age-Time	Biased ML	Debiased ML
Argentina	1996-2015	$0.990^{\dagger}$	0.955	$0.999^{\dagger}$	0.912 ± 0.107
Brazil	1993-2018	$0.994^{\dagger}$	0.980	$0.999^{\dagger}$	0.850 ± 0.240
Canada	1996-2016	0.952	0.909	0.913	0.910 ± 0.019
Denmark	1996-2016	0.991	0.979	0.984	0.989 ± 0.009
France	1996-2016	0.955	0.938	0.953	0.949 ± 0.024
Germany	2001-2016	0.999	1.012	0.995	0.996 ± 0.001
Italy	1996-2016	0.980	$0.998^{\dagger}$	0.815	0.841 ± 0.225
Mexico	2005-2019	0.740	0.937	0.778	0.853 ± 0.031
Norway	1996-2017	0.994	1.003	0.996	0.995 ± 0.002
Spain	2005-2018	1.001	0.973	1.000	0.76 ± 11.85
Sweden	1996-2016	$0.997^{\dagger}$	0.974	0.961	0.01 ± 1238
USA	1998-2019	0.934	0.878	0.882	0.854 ± 0.075

Note: The table presents results on the persistence of income shocks—ρ in Section 3—for the twelve countries we have data for in GRID. Items with † indicate that the slope is negative. See the note for Table 1 for more detail.

I Additional Results for Average Income

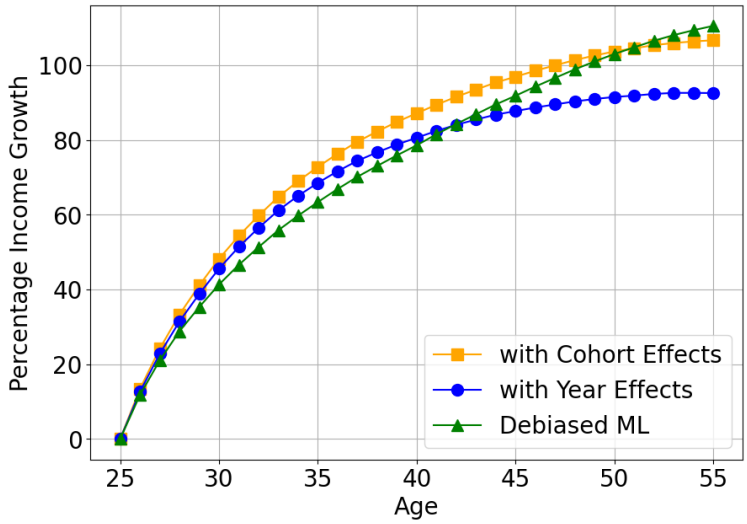


Figure I.1: Age Profile of Income Growth in the US (GRID)

Note: The figure plots the estimated age profile of log income in levels for three models using GRID data. All three profiles are shifted to begin at zero, and then multiplied by 100 to give percentage growth.